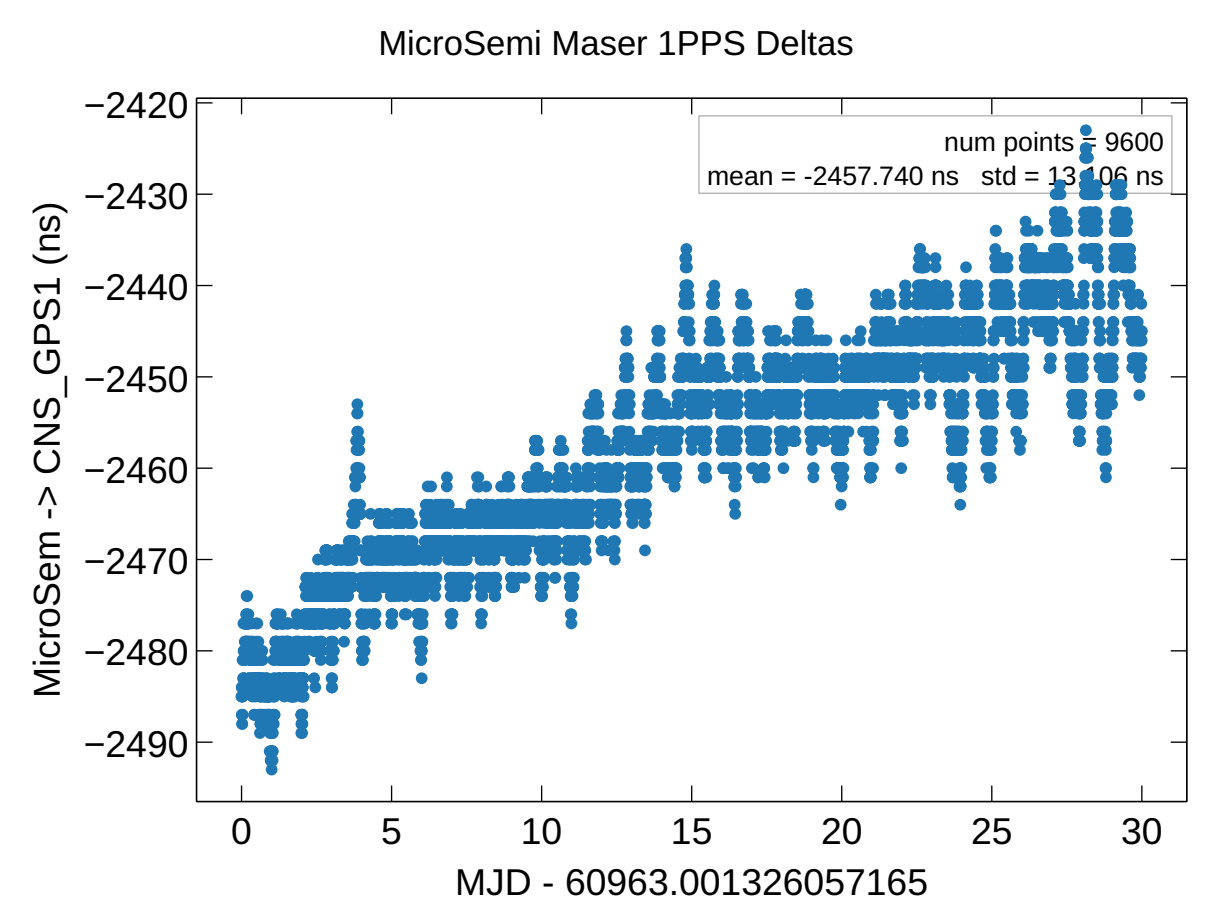
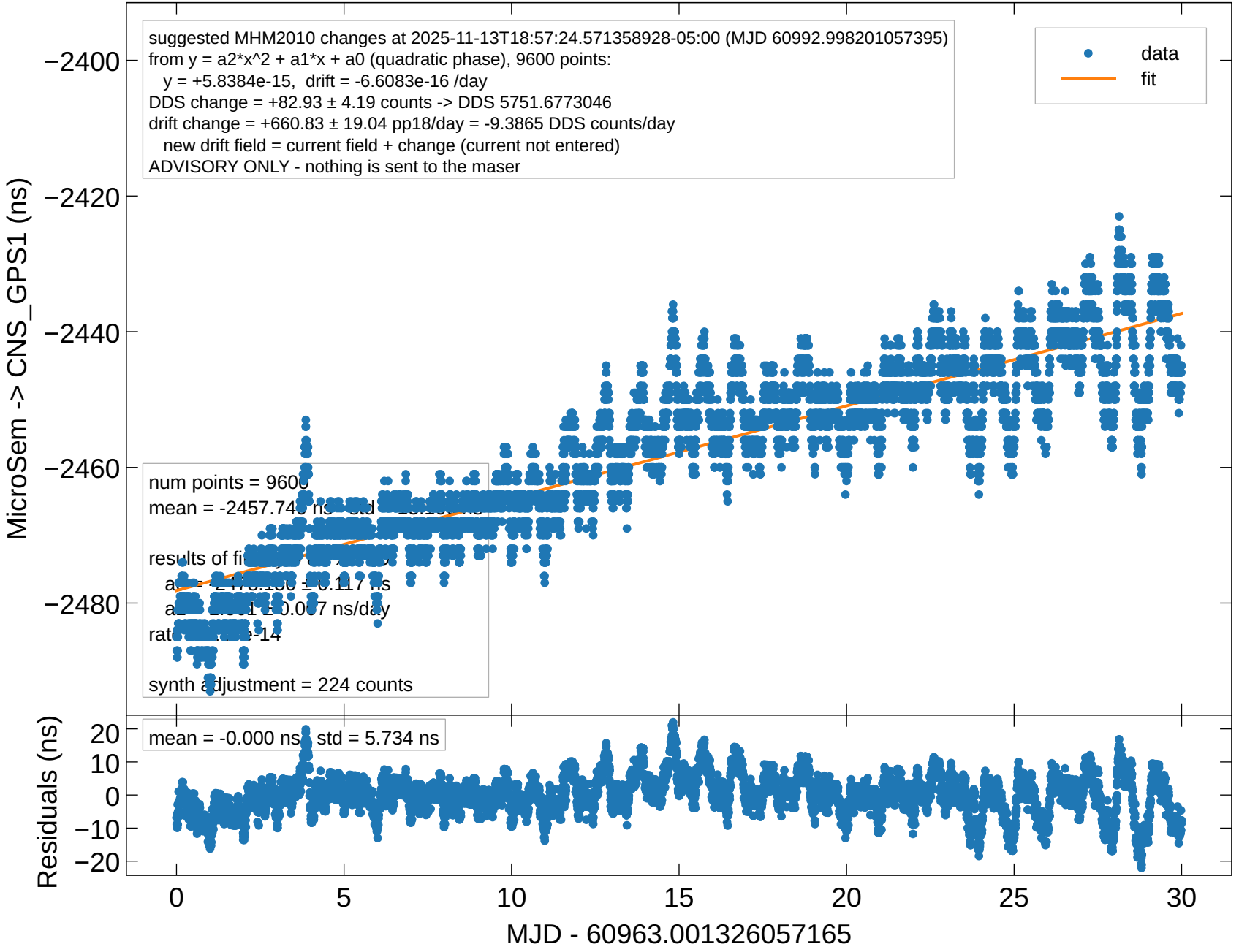
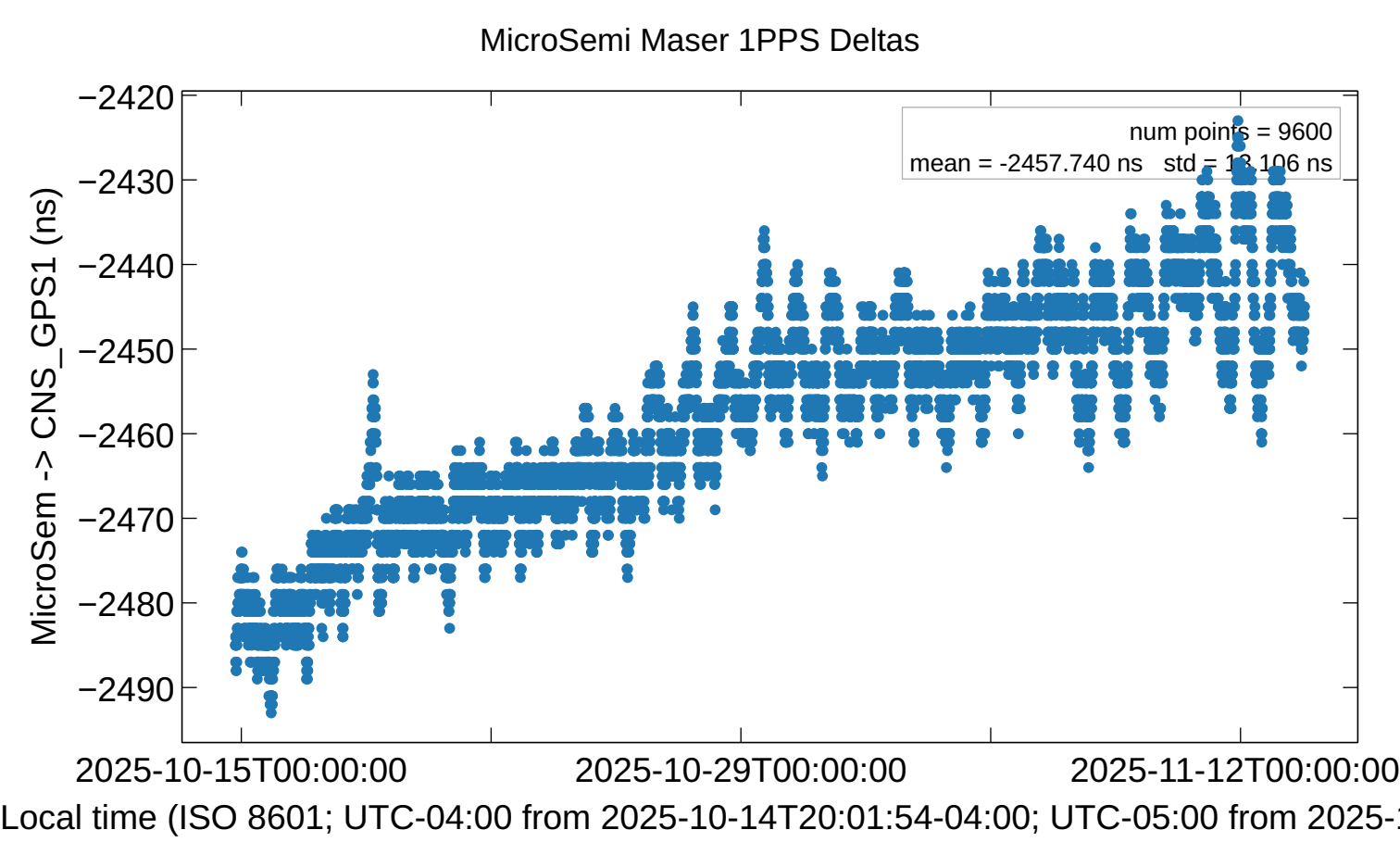
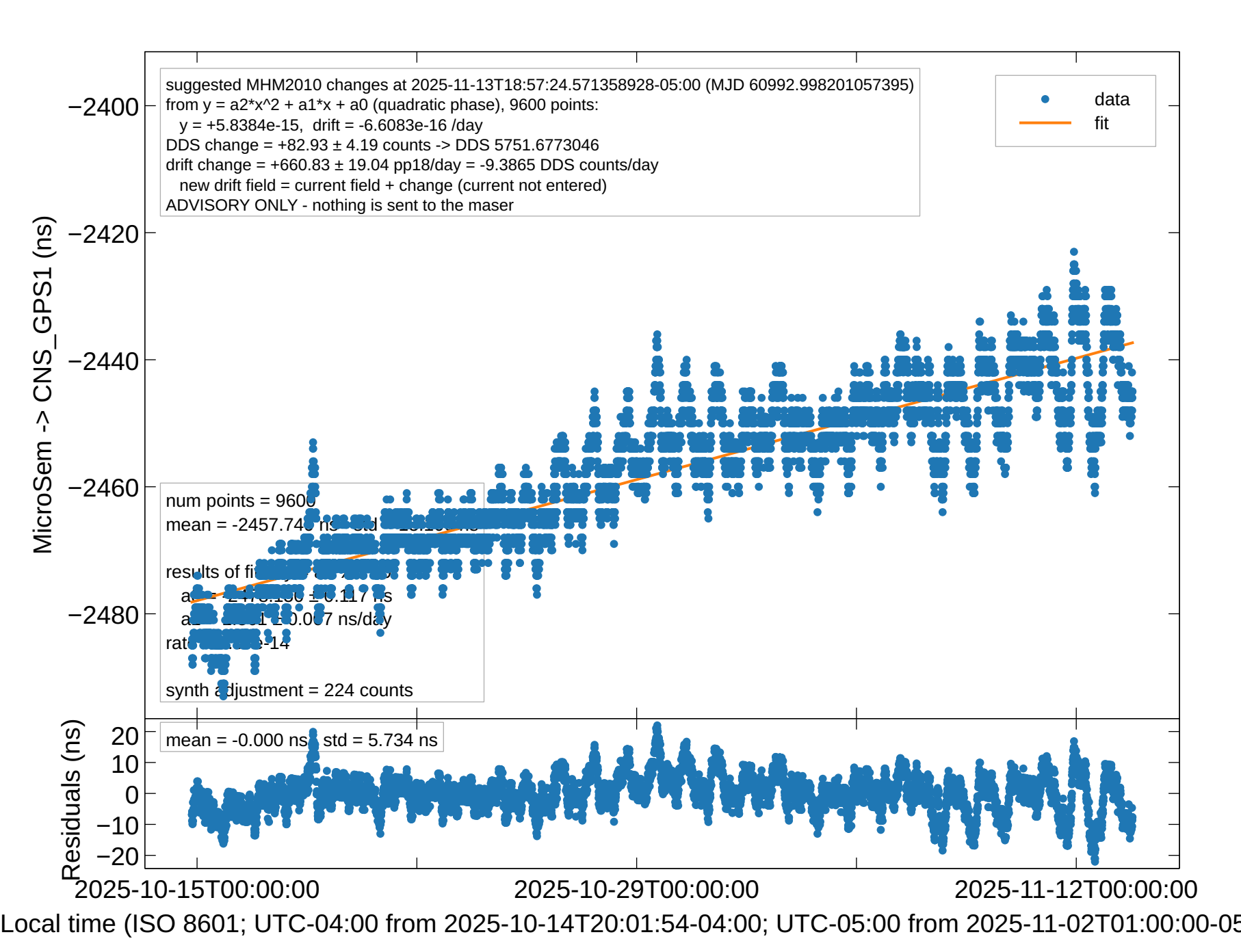
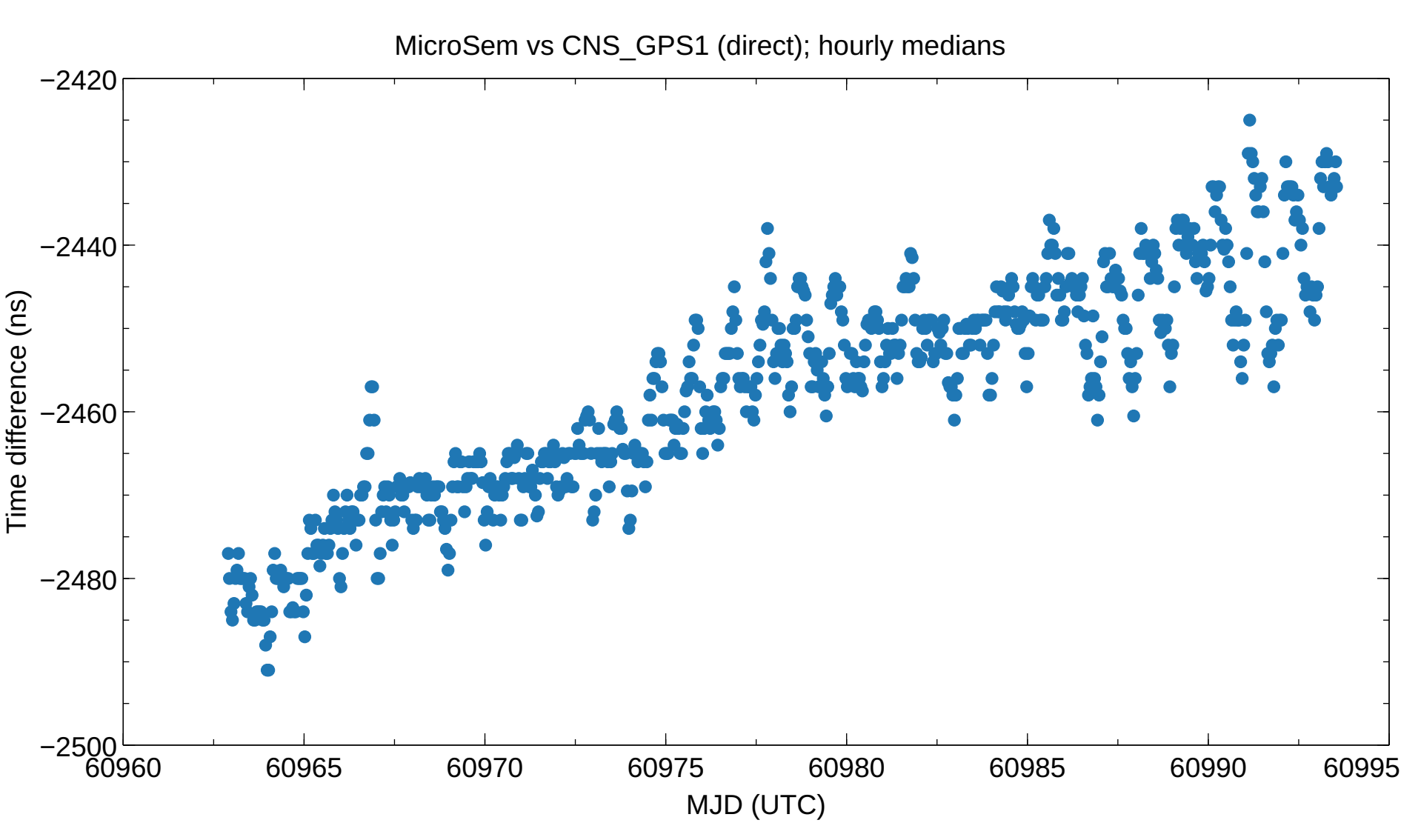
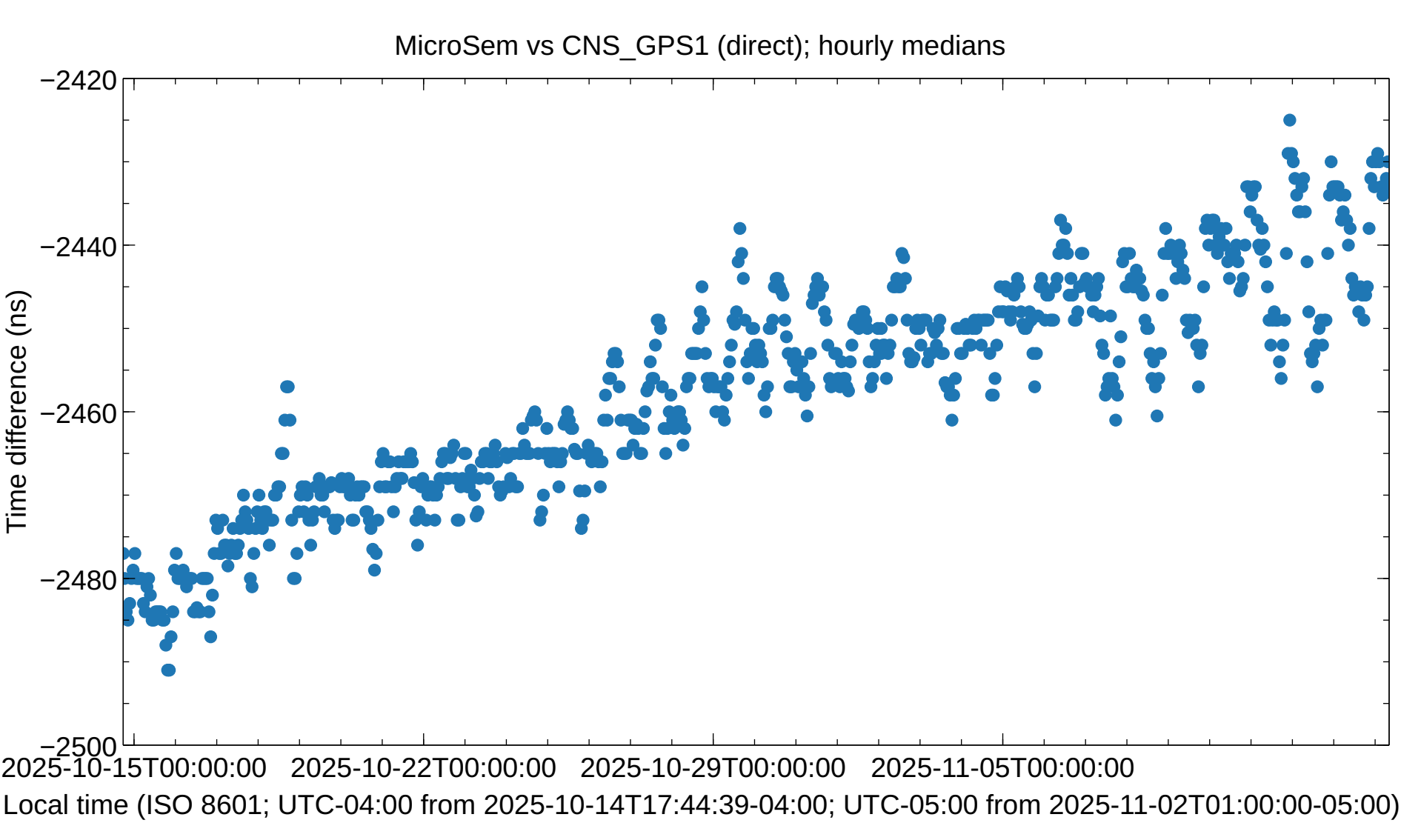

MicroSemi Maser 1PPS Deltas
num points = 9600
mean = -2457.740 ns std = 13.106 ns
MicroSem -> CNS_GPS1 (ns)
−2420
−2430
−2440
−2450
−2460
−2470
−2480
−2490
0
5
10
15
20
25
30
MJD - 60963.001326057165

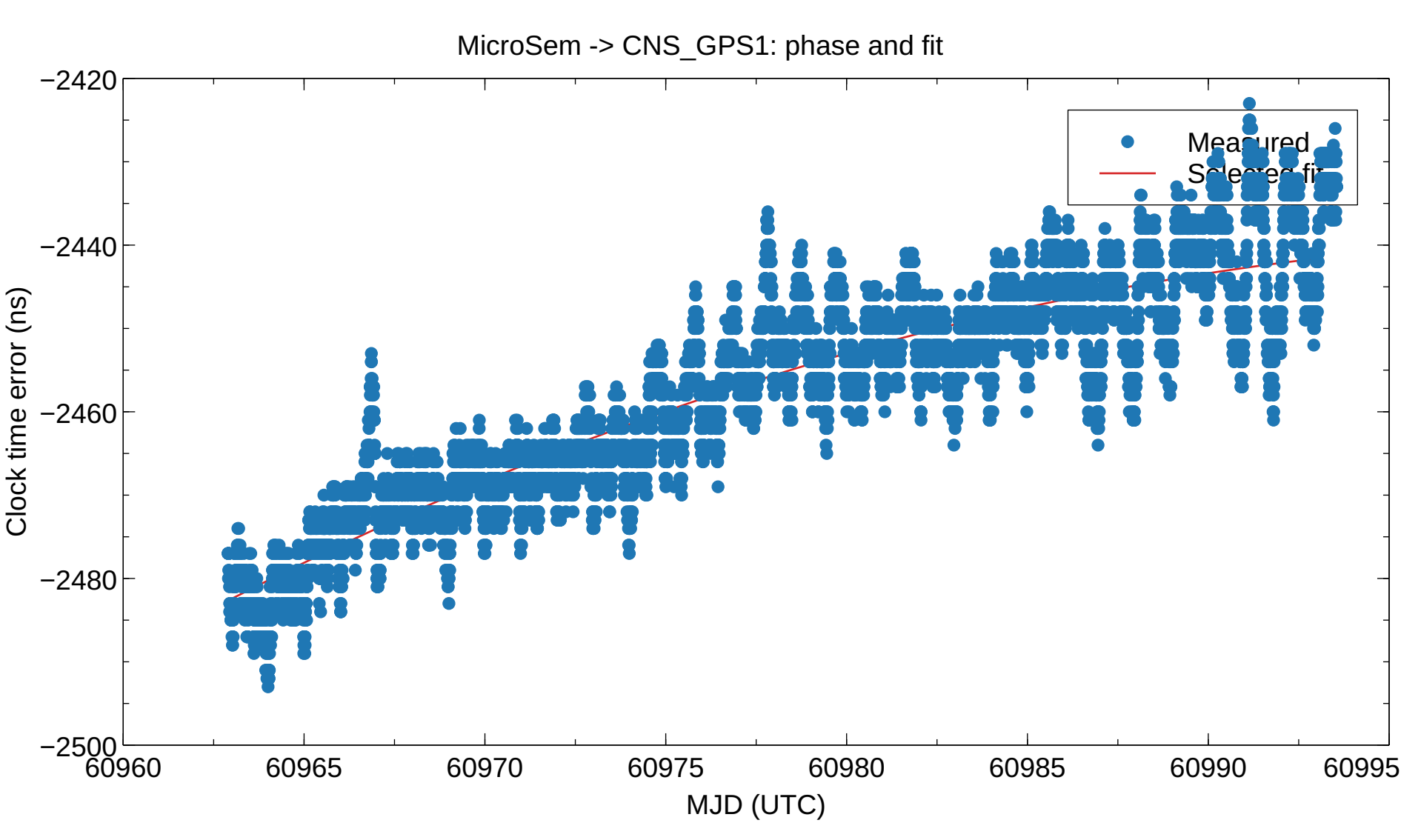

suggested MHM2010 changes at 2025-11-13T18:57:24.571358928-05:00 (MJD 60992.998201057395)
from y = a2*x^2 + a1*x + a0 (quadratic phase), 9600 points:
y = +5.8384e-15, drift = -6.6083e-16 /day
DDS change = +82.93 ± 4.19 counts -> DDS 5751.6773046
drift change = +660.83 ± 19.04 pp18/day = -9.3865 DDS counts/day
new drift field = current field + change (current not entered)
ADVISORY ONLY - nothing is sent to the maser
data
fit
num points = 9600
synth adjustment = 224 counts
mean = -0.000 ns, std = 5.734 ns
MicroSem -> CNS_GPS1 (ns)
Residuals (ns)
MJD - 60963.001326057165

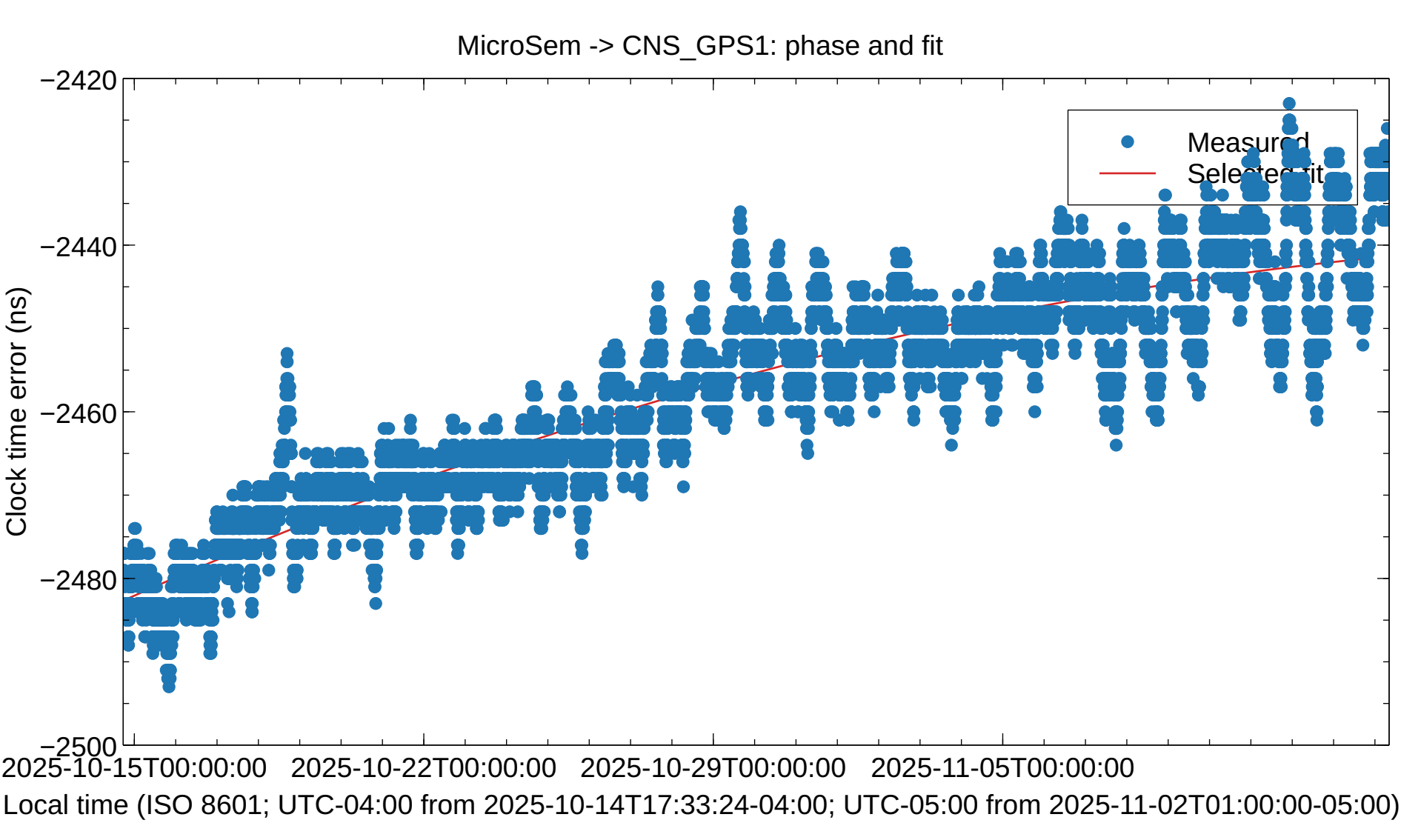

MicroSemi Maser 1PPS Deltas
num points = 9600
mean = -2457.740 ns std = 13.106 ns
MicroSem -> CNS_GPS1 (ns)
−2420
−2430
−2440
−2450
−2460
−2470
−2480
−2490
2025-10-15T00:00:00
2025-10-29T00:00:00
2025-11-12T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:01:54-04:00; UTC-05:00 from 2025-

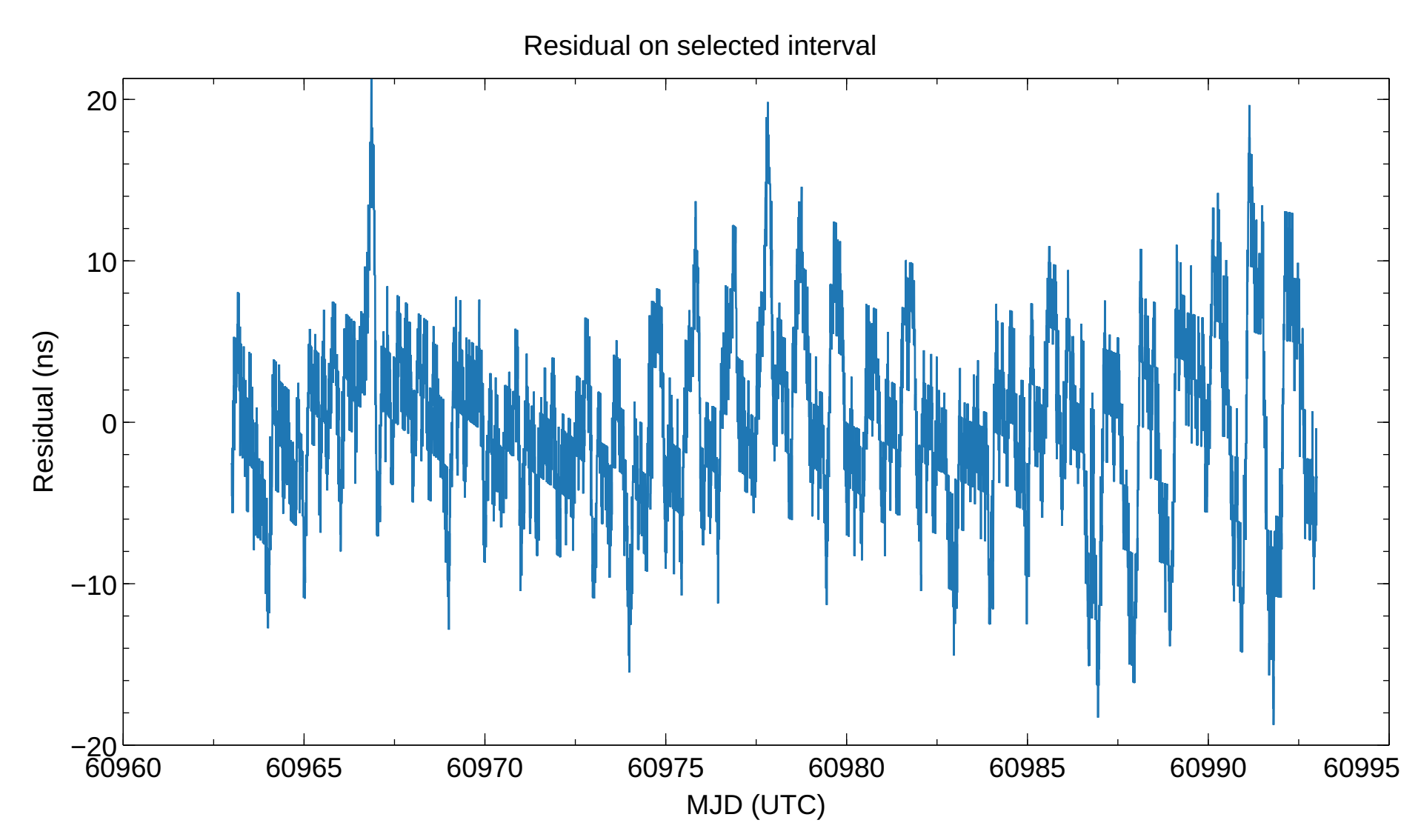

suggested MHM2010 changes at 2025-11-13T18:57:24.571358928-05:00 (MJD 60992.998201057395)
from y = a2*x^2 + a1*x + a0 (quadratic phase), 9600 points:
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drift change = +660.83 ± 19.04 pp18/day = -9.3865 DDS counts/day
new drift field = current field + change (current not entered)
ADVISORY ONLY - nothing is sent to the maser
data
fit
num points = 9600
synth adjustment = 224 counts
mean = -0.000 ns, std = 5.734 ns
MicroSem -> CNS_GPS1 (ns)
Residuals (ns)
2025-10-15T00:00:00
2025-10-29T00:00:00
2025-11-12T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:01:54-04:00; UTC-05:00 from 2025-11-02T01:00:00-05

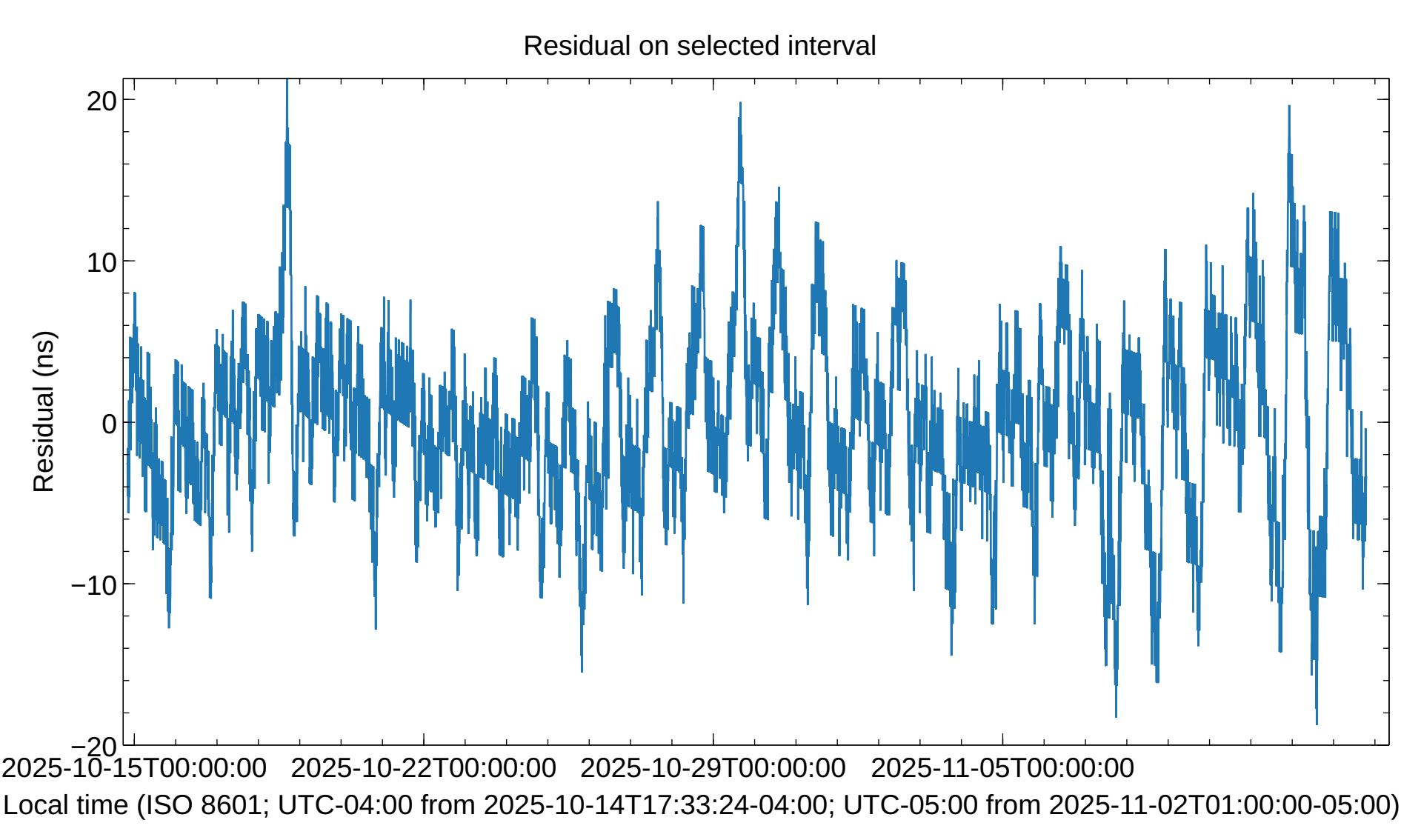

MicroSem vs CNS_GPS1 (direct); hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

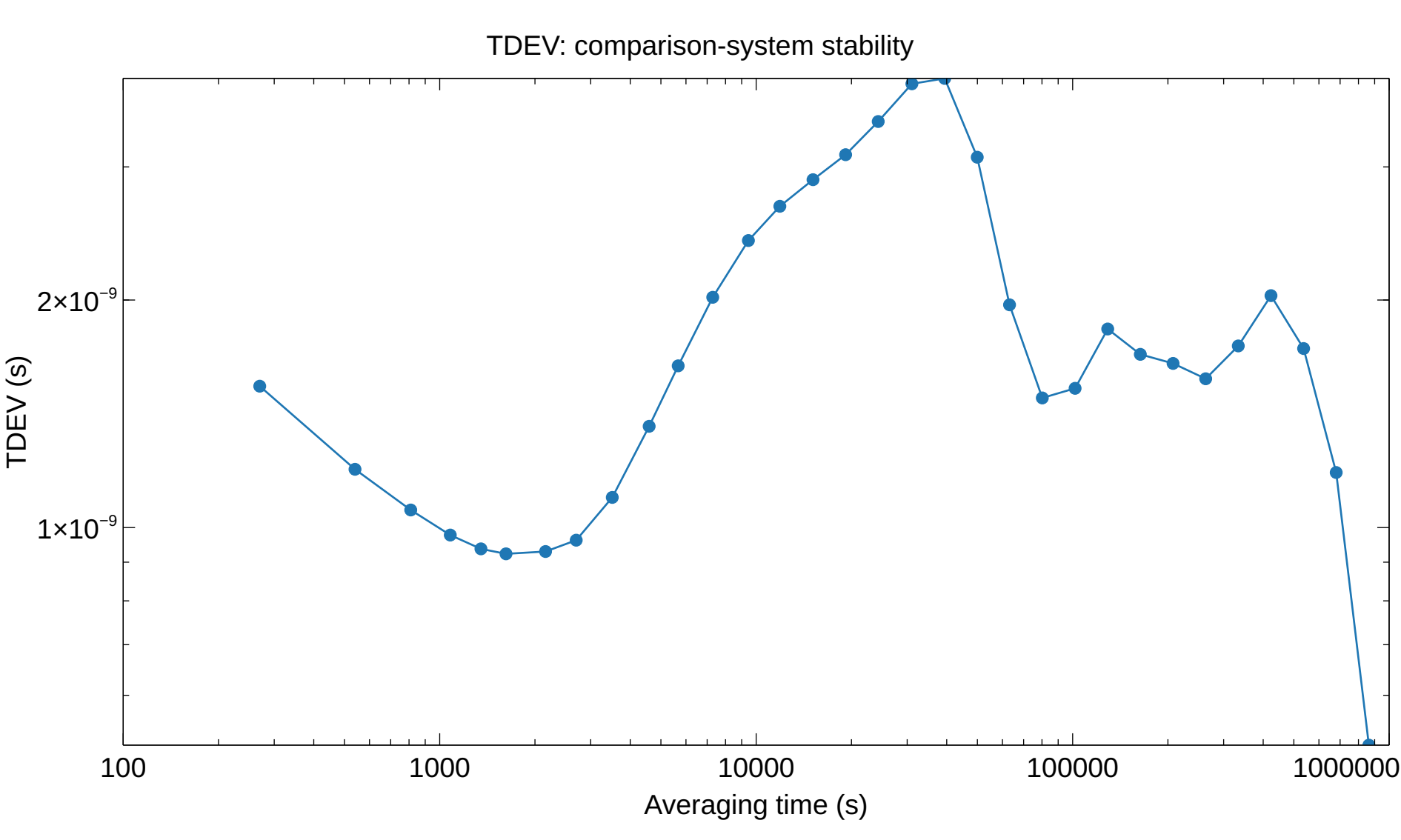

MicroSem vs CNS_GPS1 (direct); hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T17:44:39-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)

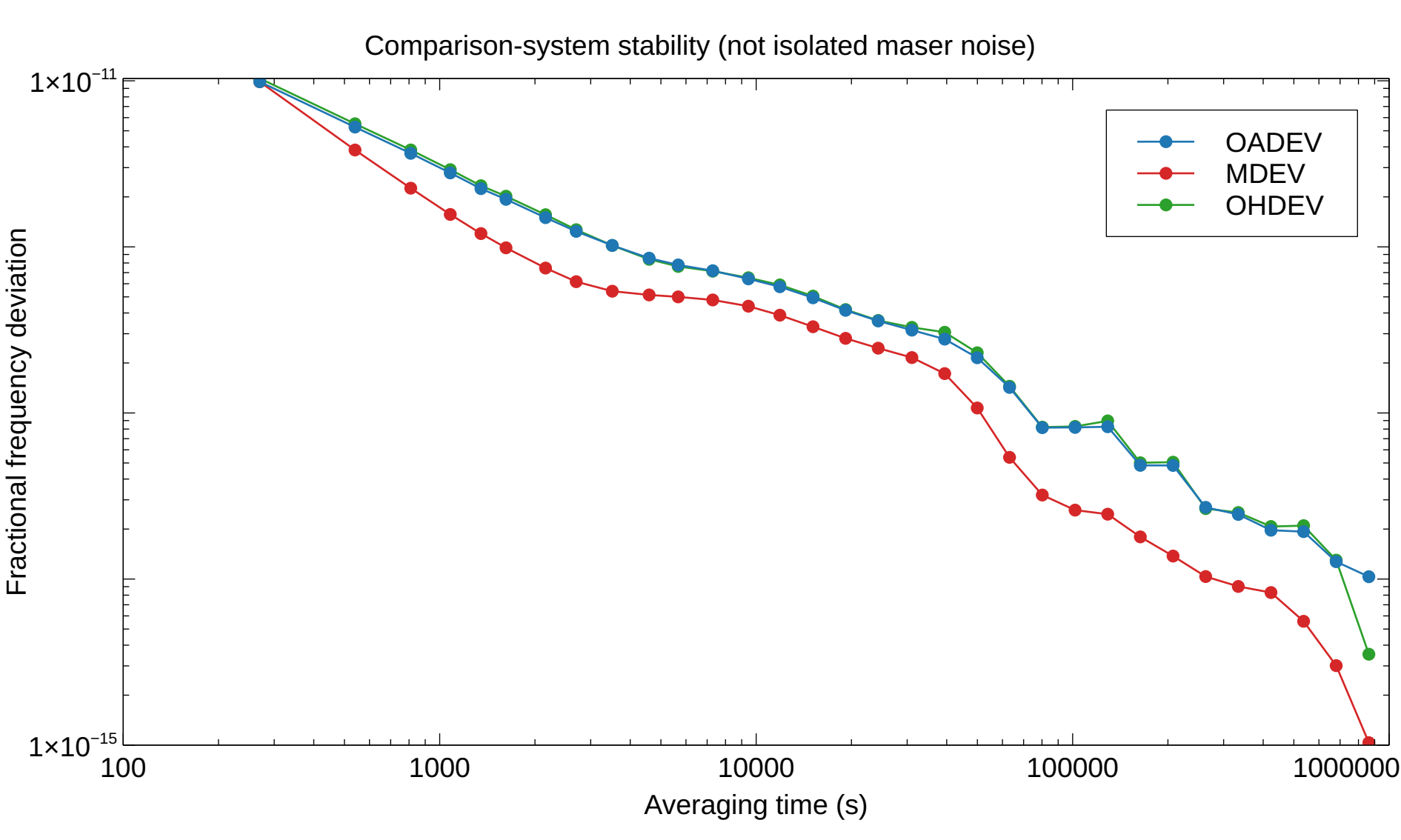

MicroSem -> CNS_GPS1: phase and fit
Measured
Selected fit
Clock time error (ns)
−2420
−2440
−2460
−2480
−2500
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

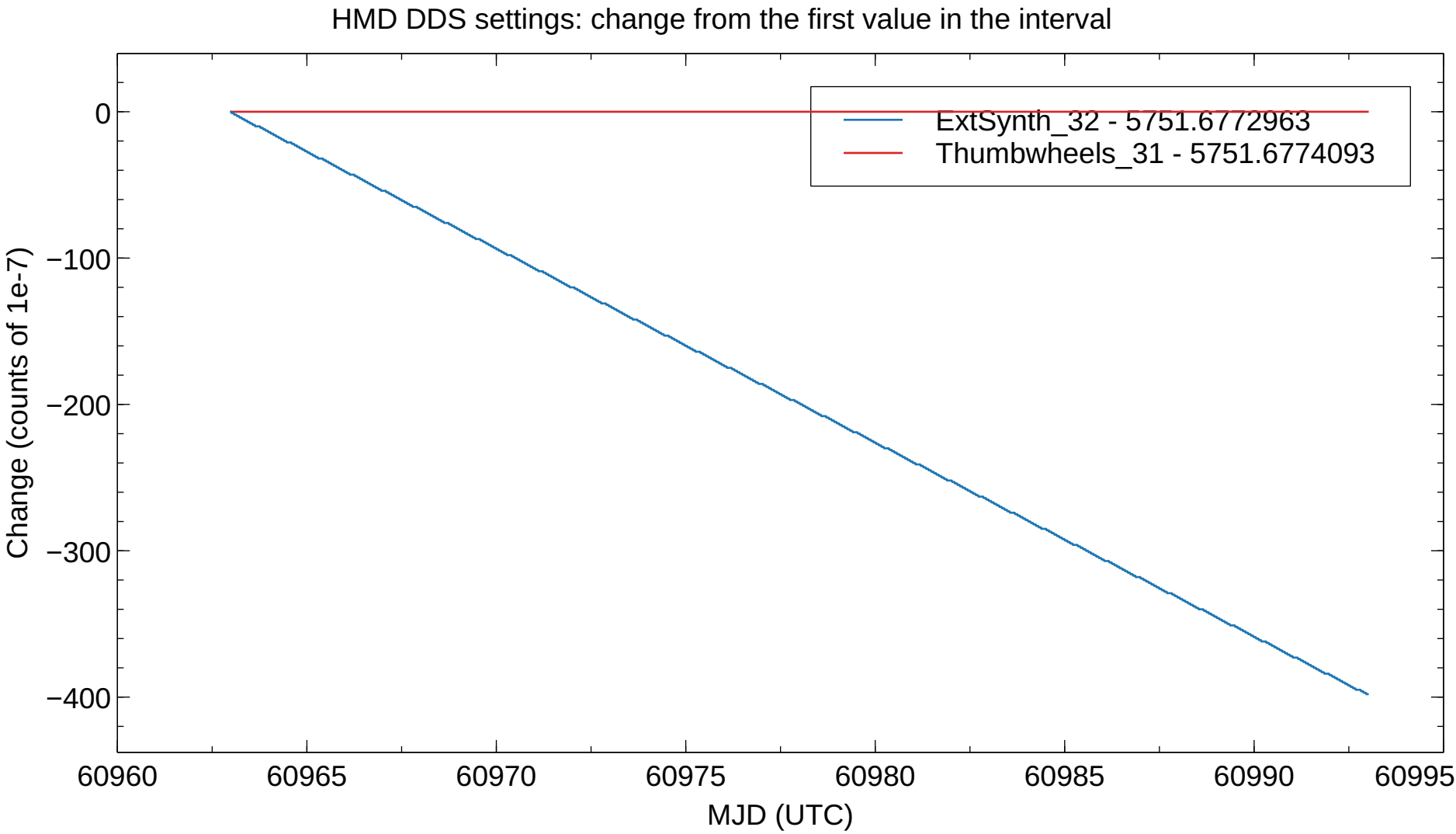

MicroSem -> CNS_GPS1: phase and fit
Measured
Selected fit
Clock time error (ns)
−2420
−2440
−2460
−2480
−2500
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T17:33:24-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)

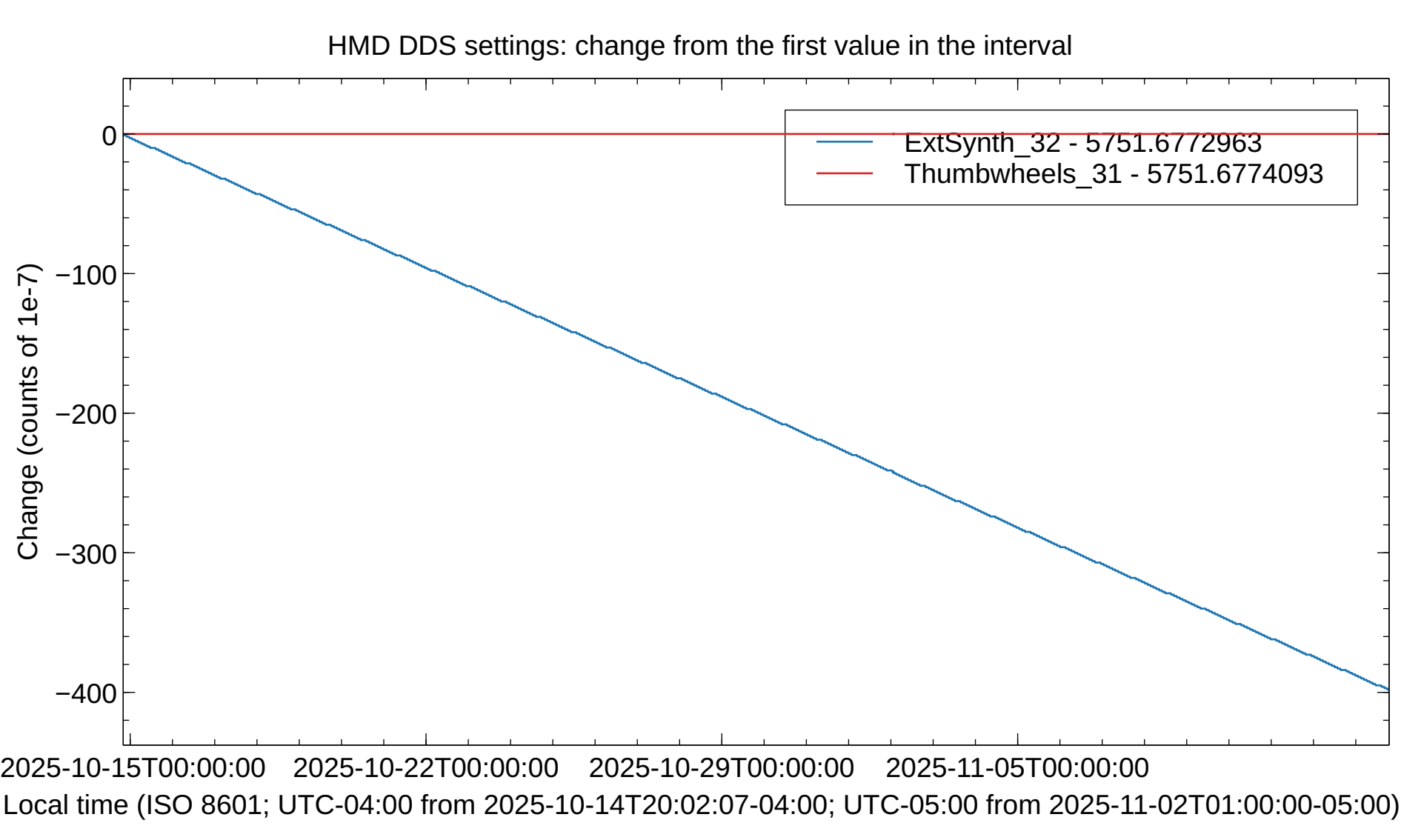

Residual on selected interval
Residual (ns)
20
10
0
−10
−20
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

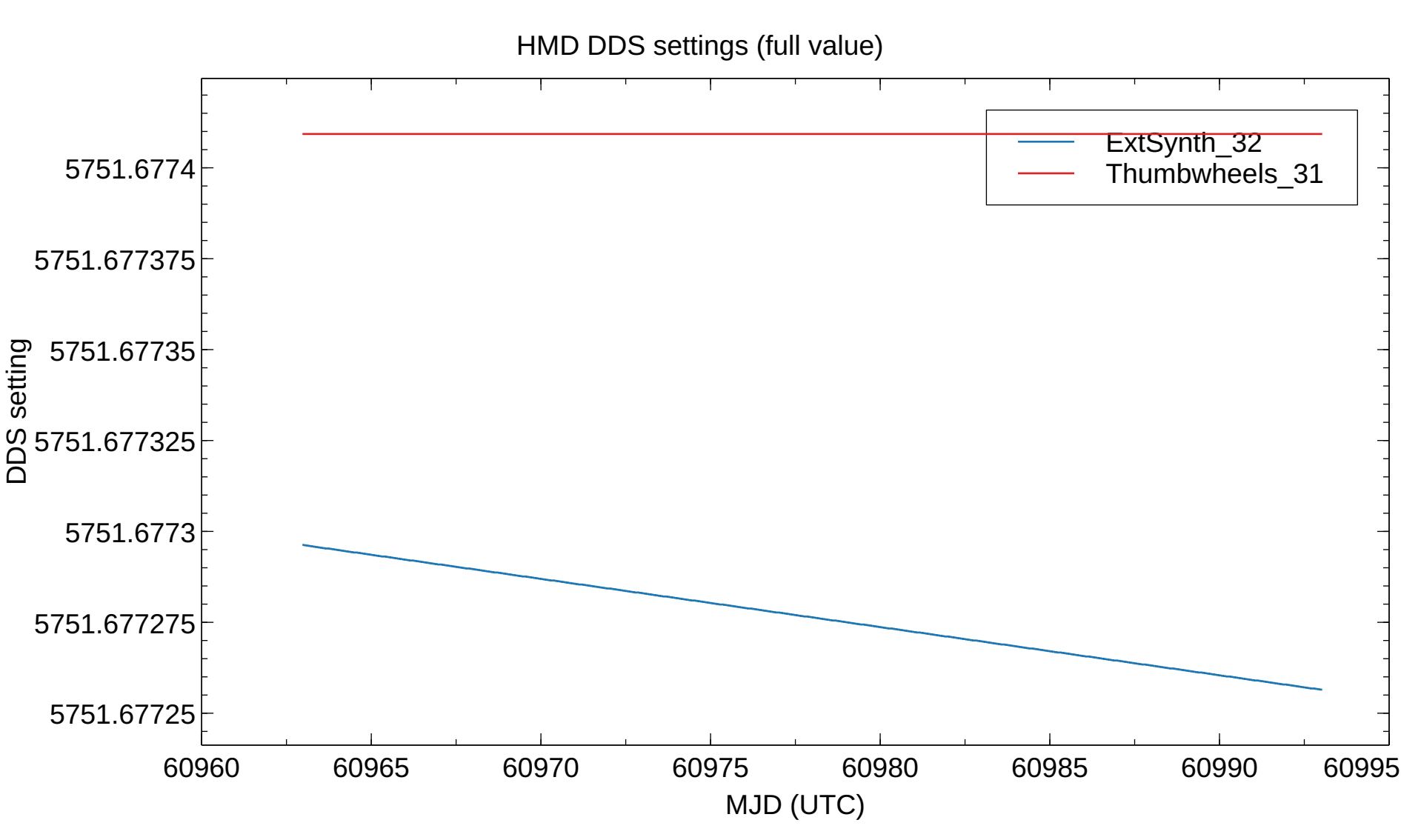

Residual on selected interval
Residual (ns)
20
10
0
−10
−20
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T17:33:24-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)

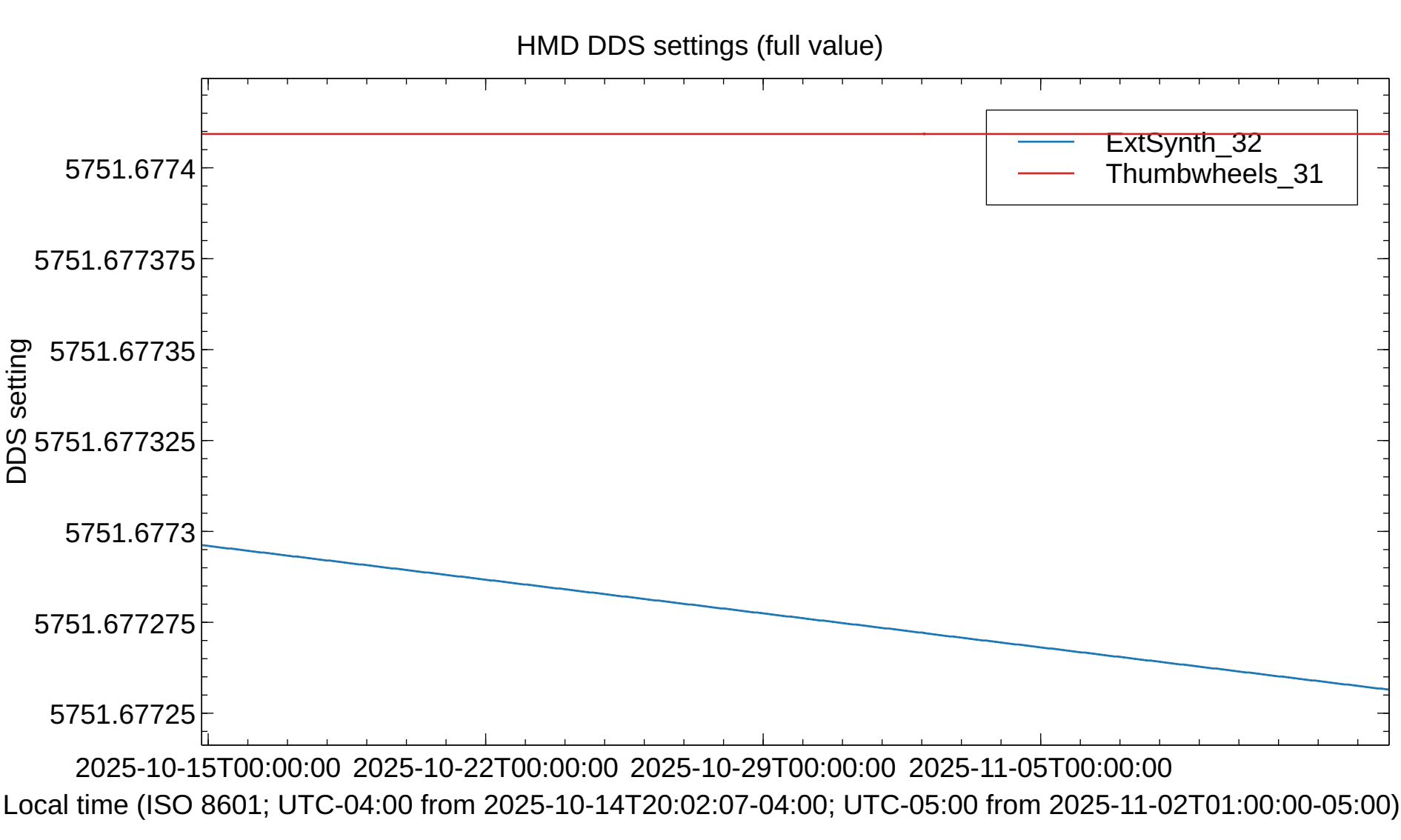

TDEV: comparison-system stability
TDEV (s)
2×10⁻⁹
1×10⁻⁹
100
1000
10000
100000
1000000
Averaging time (s)

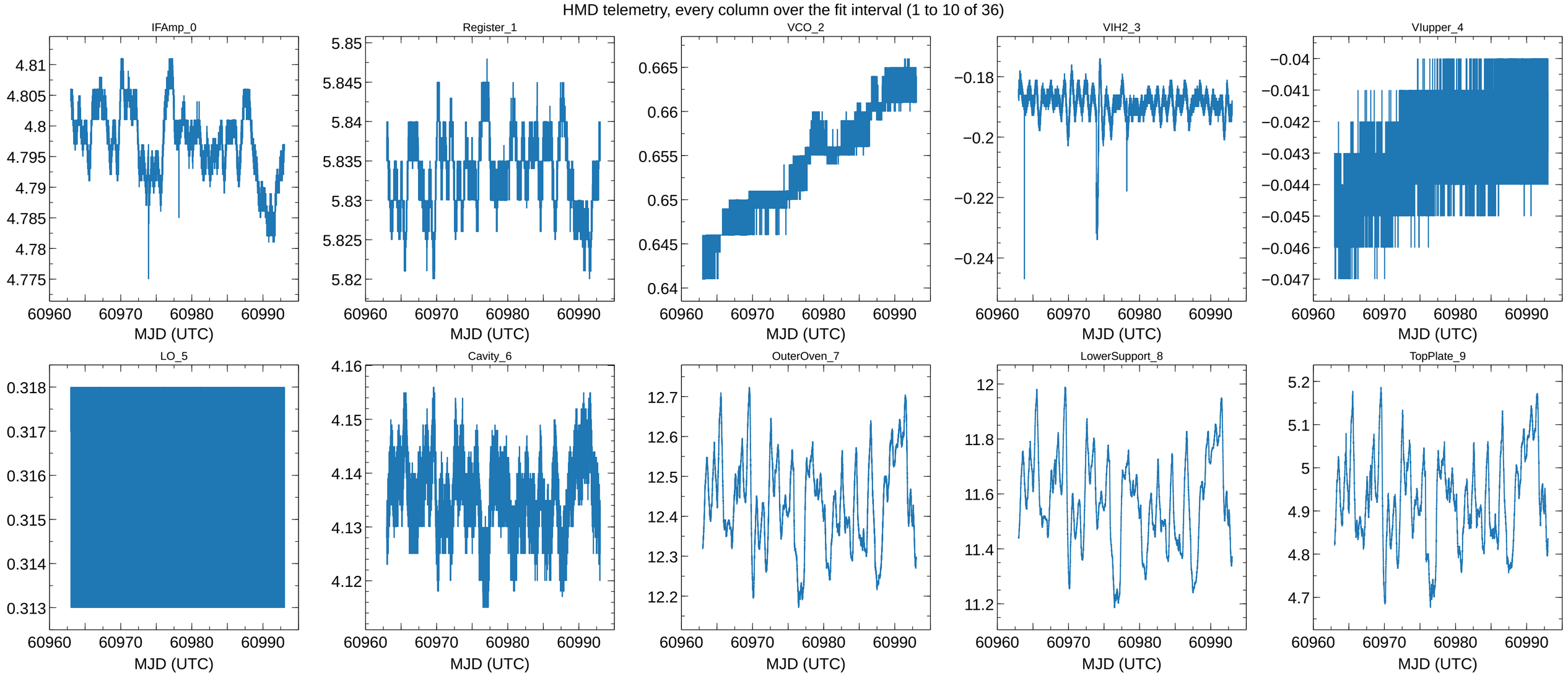

Comparison-system stability (not isolated maser noise)
1×10−11
1×10−15
Fractional frequency deviation
100
1000
10000
100000
1000000
Averaging time (s)
OADEV
MDEV
OHDEV


HMD DDS settings: change from the first value in the interval
ExtSynth_32 - 5751.6772963
Thumbwheels_31 - 5751.6774093
0
−100
−200
−300
−400
Change (counts of 1e-7)
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

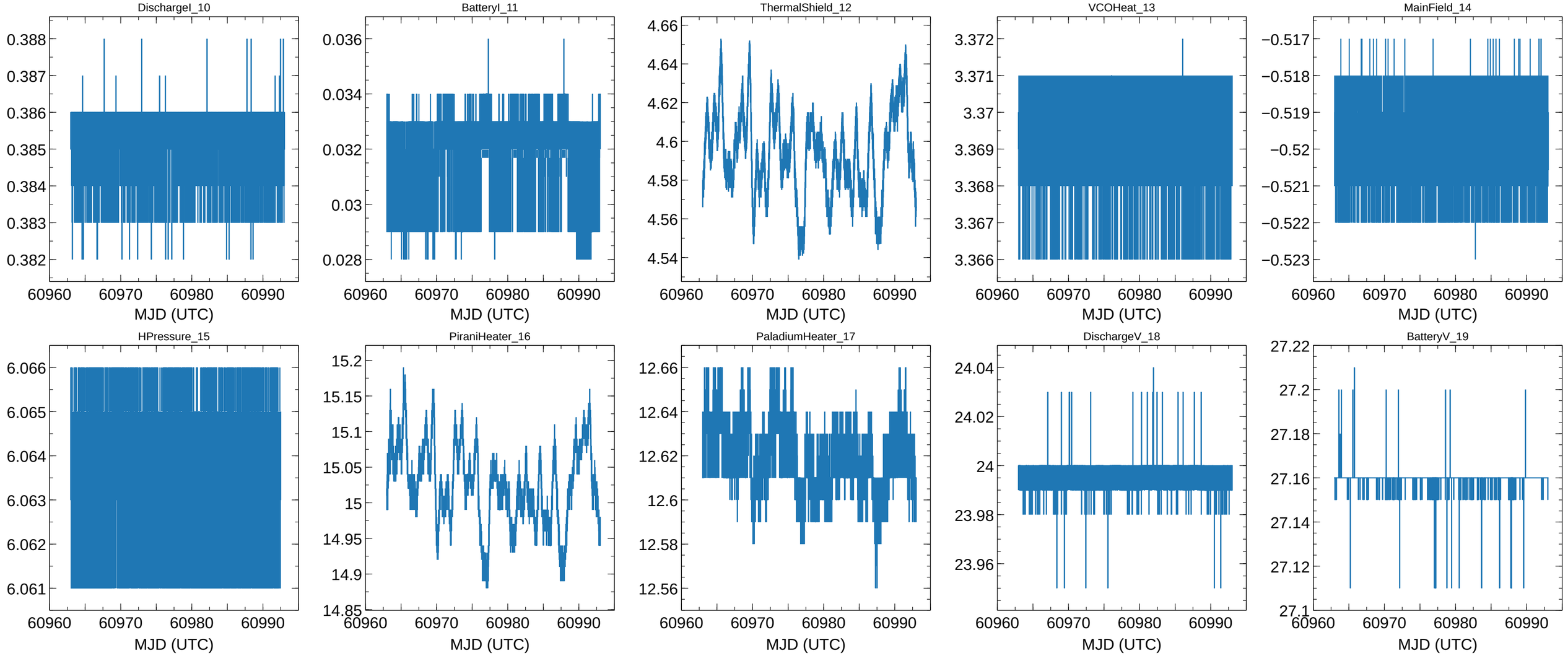

HMD DDS settings: change from the first value in the interval
ExtSynth_32 - 5751.6772963
Thumbwheels_31 - 5751.6774093
0
−100
−200
−300
−400
Change (counts of 1e-7)
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:02:07-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)


HMD DDS settings (full value)
ExtSynth_32
Thumbwheels_31
DDS setting
5751.6774
5751.677375
5751.67735
5751.677325
5751.6773
5751.677275
5751.67725
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

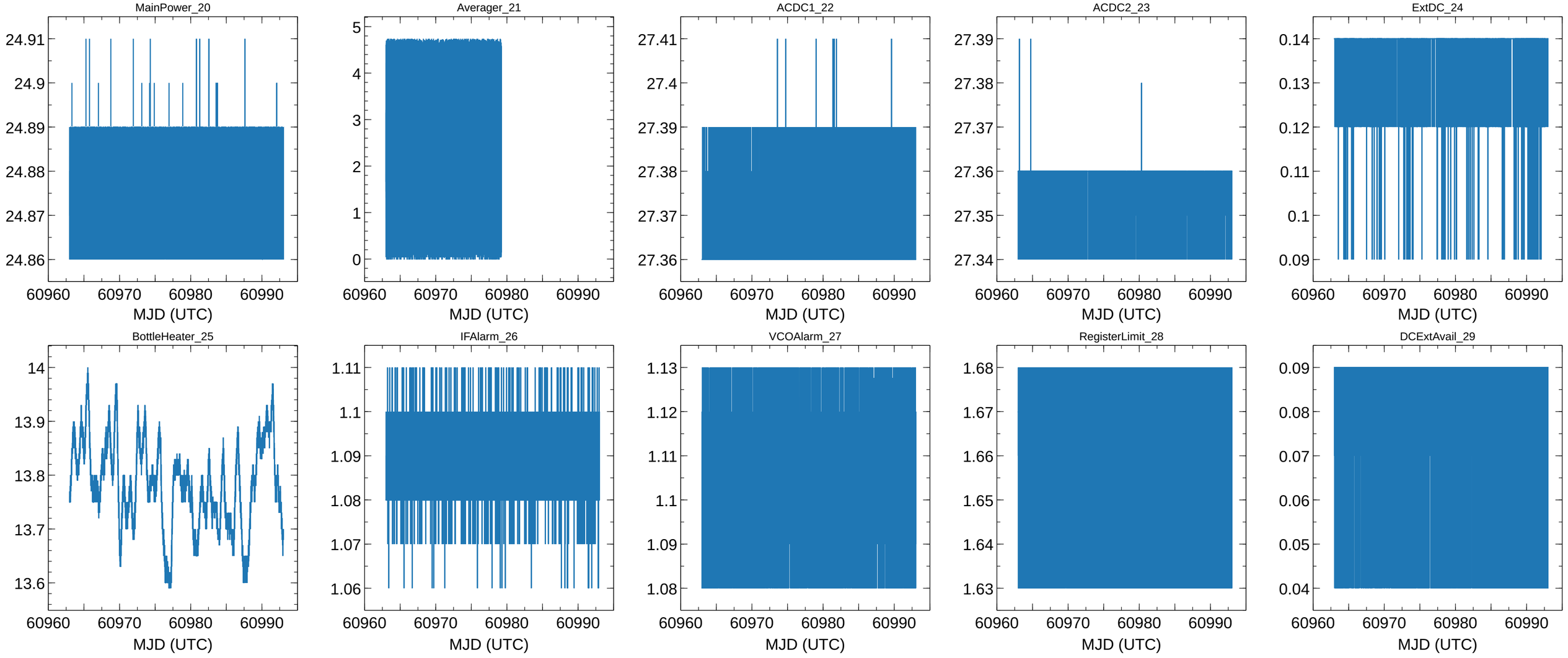

HMD DDS settings (full value)
ExtSynth_32
Thumbwheels_31
DDS setting
5751.6774
5751.677375
5751.67735
5751.677325
5751.6773
5751.677275
5751.67725
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:02:07-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)


HMD telemetry, every column over the fit interval (1 to 10 of 36)
IFAmp_0
Register_1
VCO_2
VIH2_3
VIupper_4
LO_5
Cavity_6
OuterOven_7
LowerSupport_8
TopPlate_9
MJD (UTC)

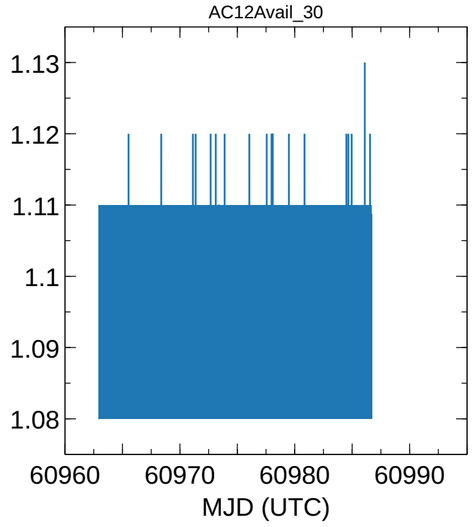
HMD telemetry, every column over the fit interval (11 to 20 of 36)

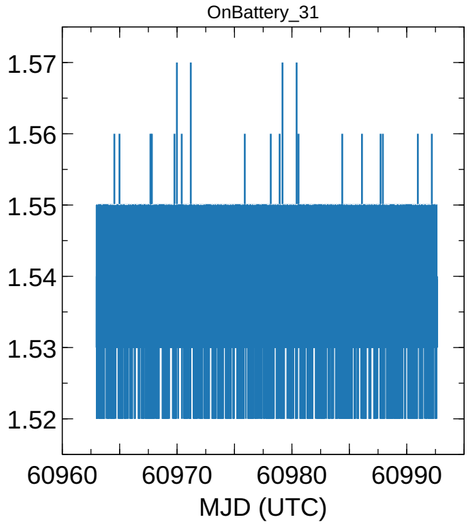


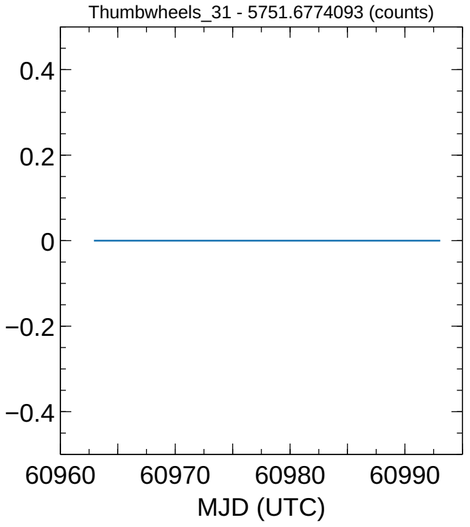
HMD telemetry, every column over the fit interval (21 to 30 of 36)

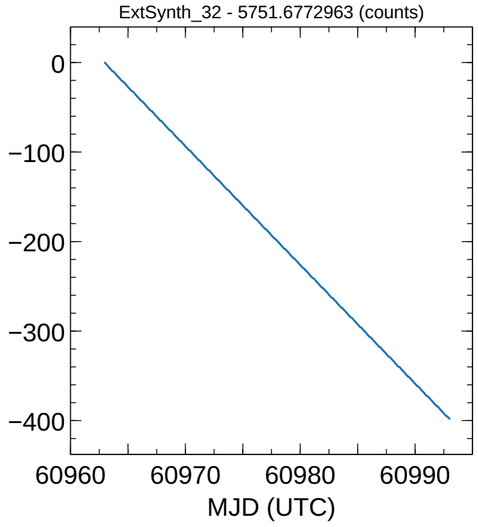


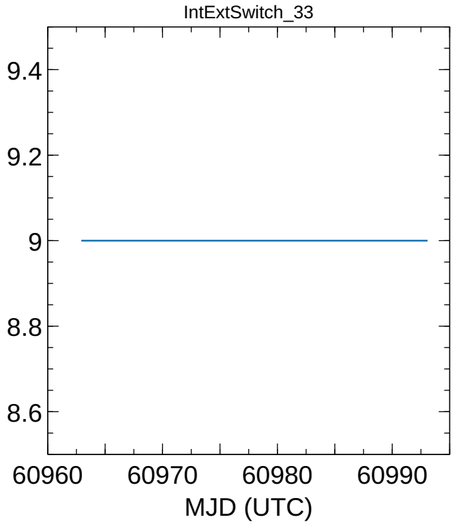
HMD telemetry, every column over the fit interval (31 to 36 of 36)

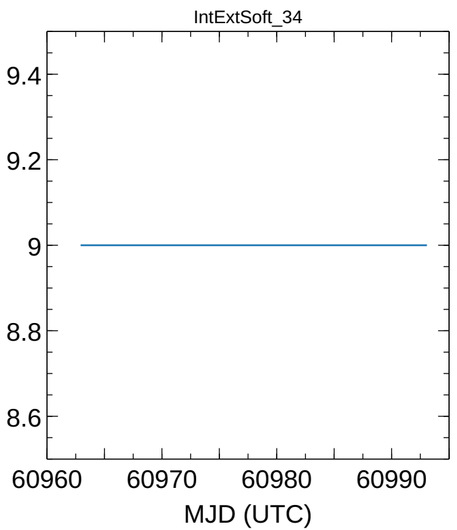


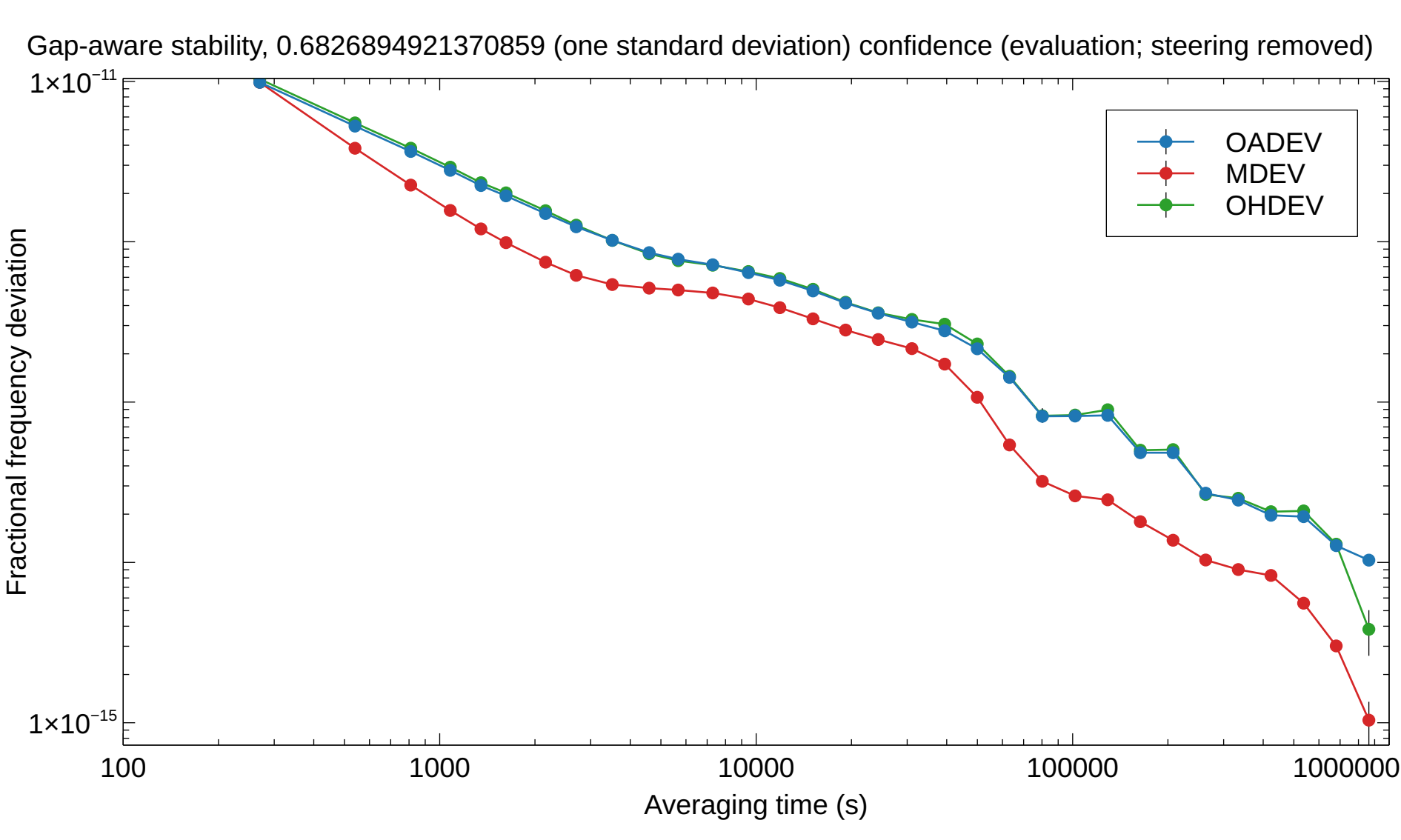


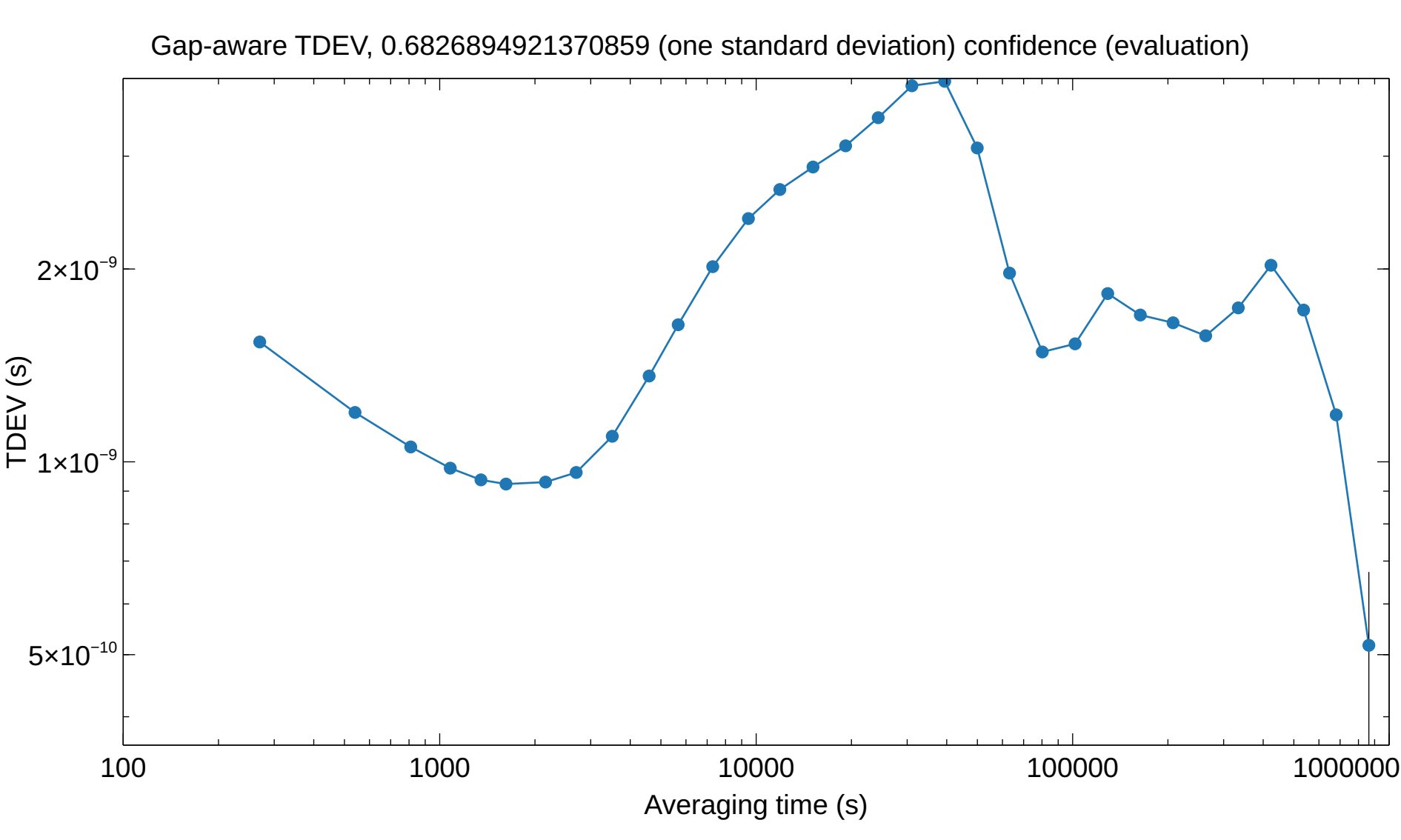


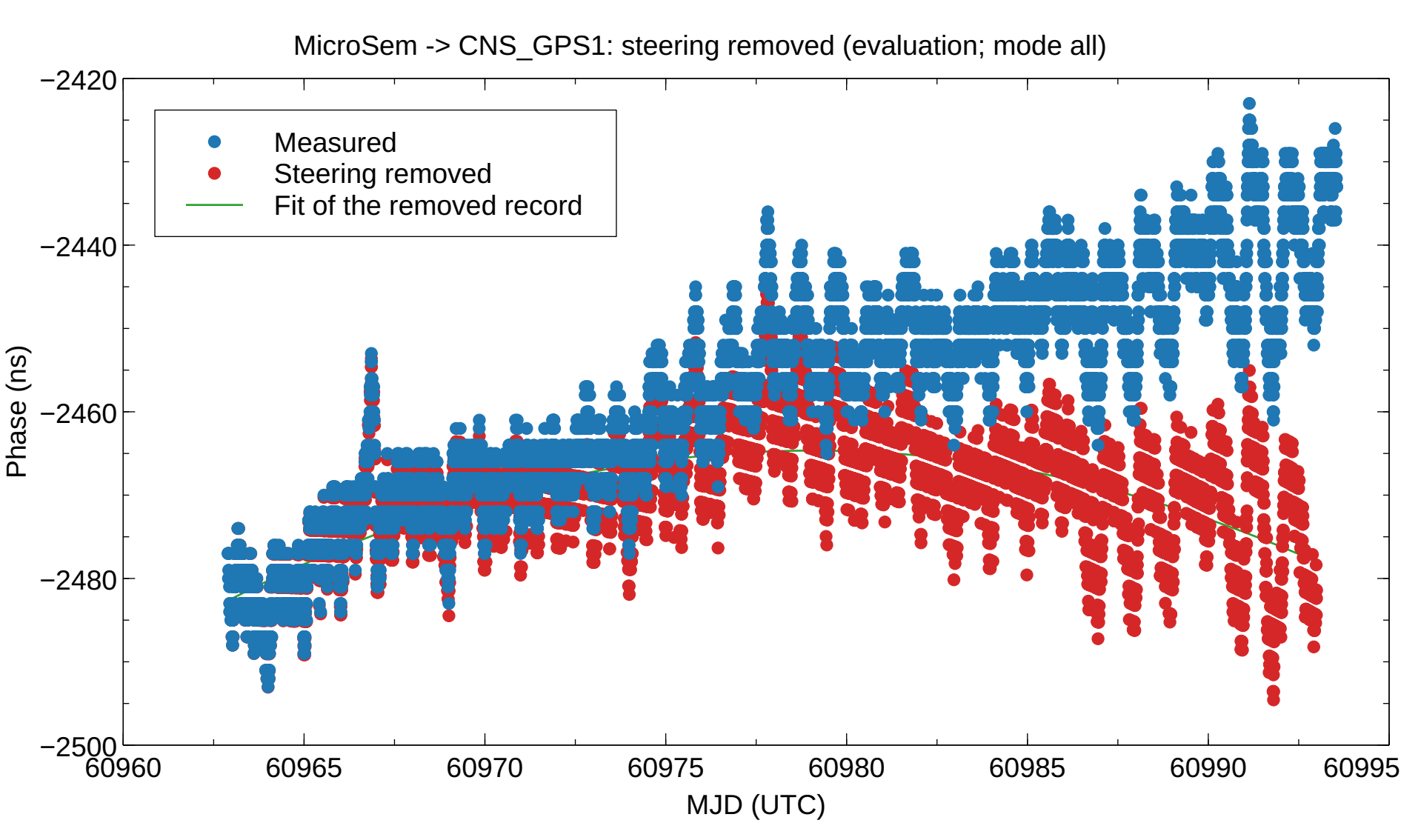


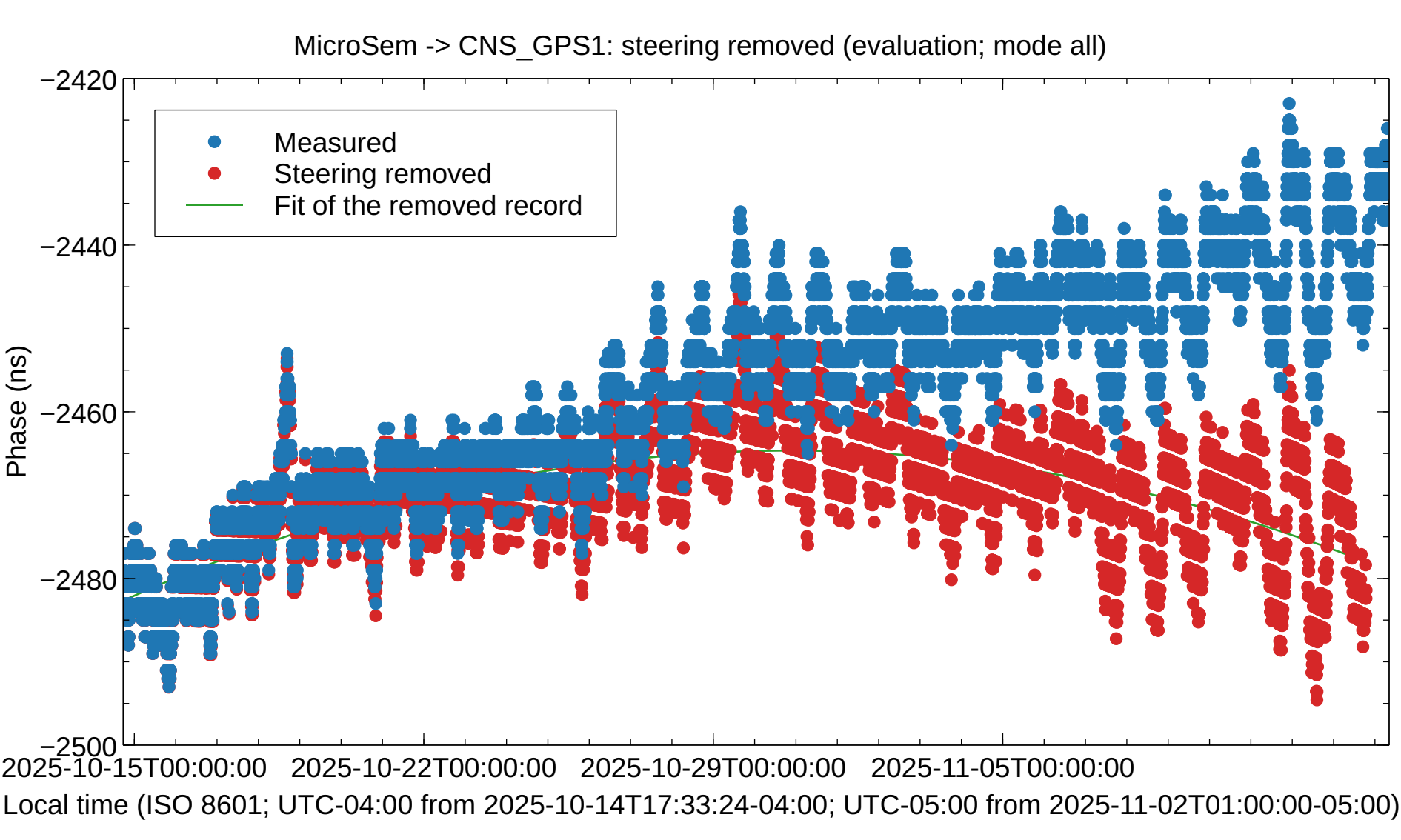


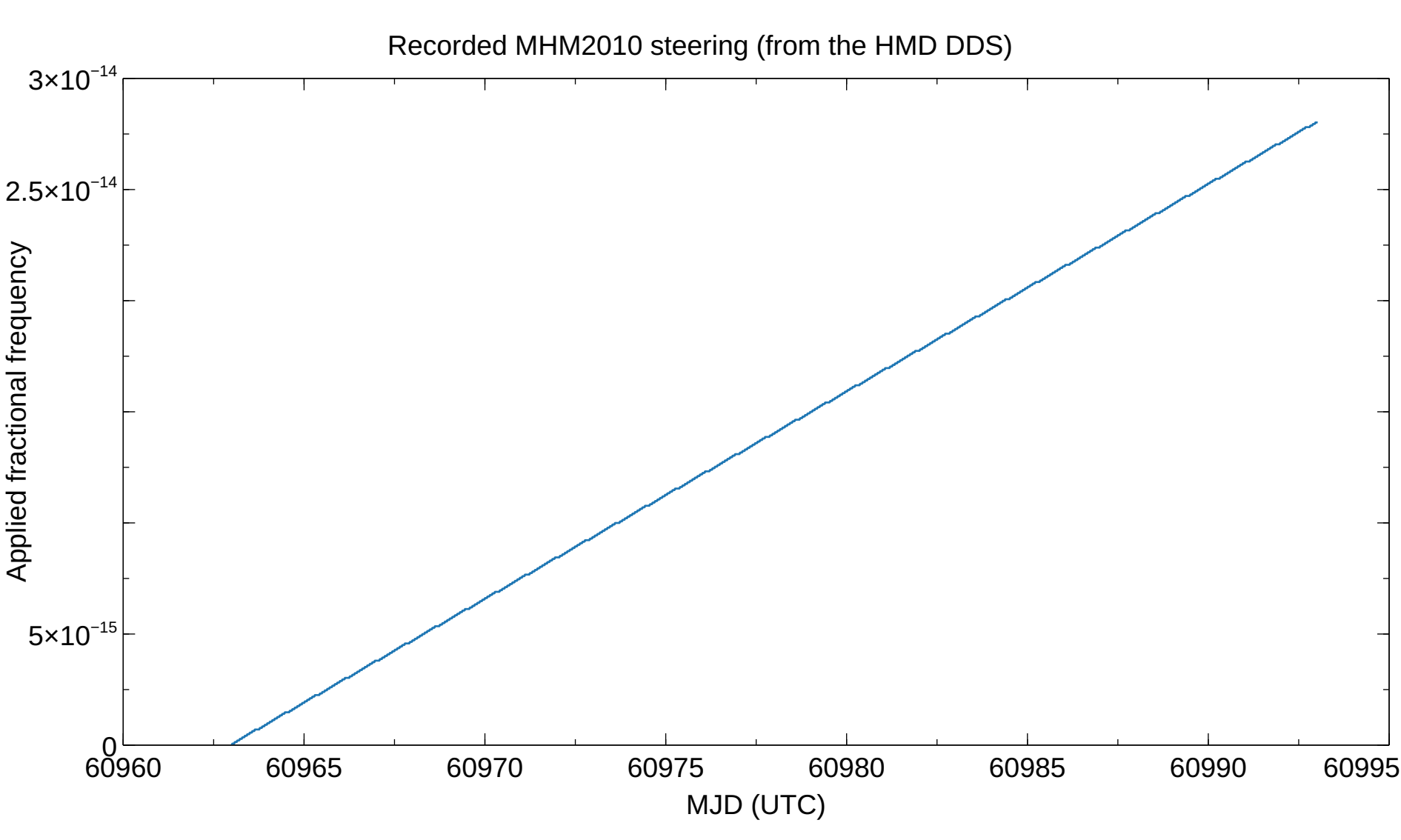


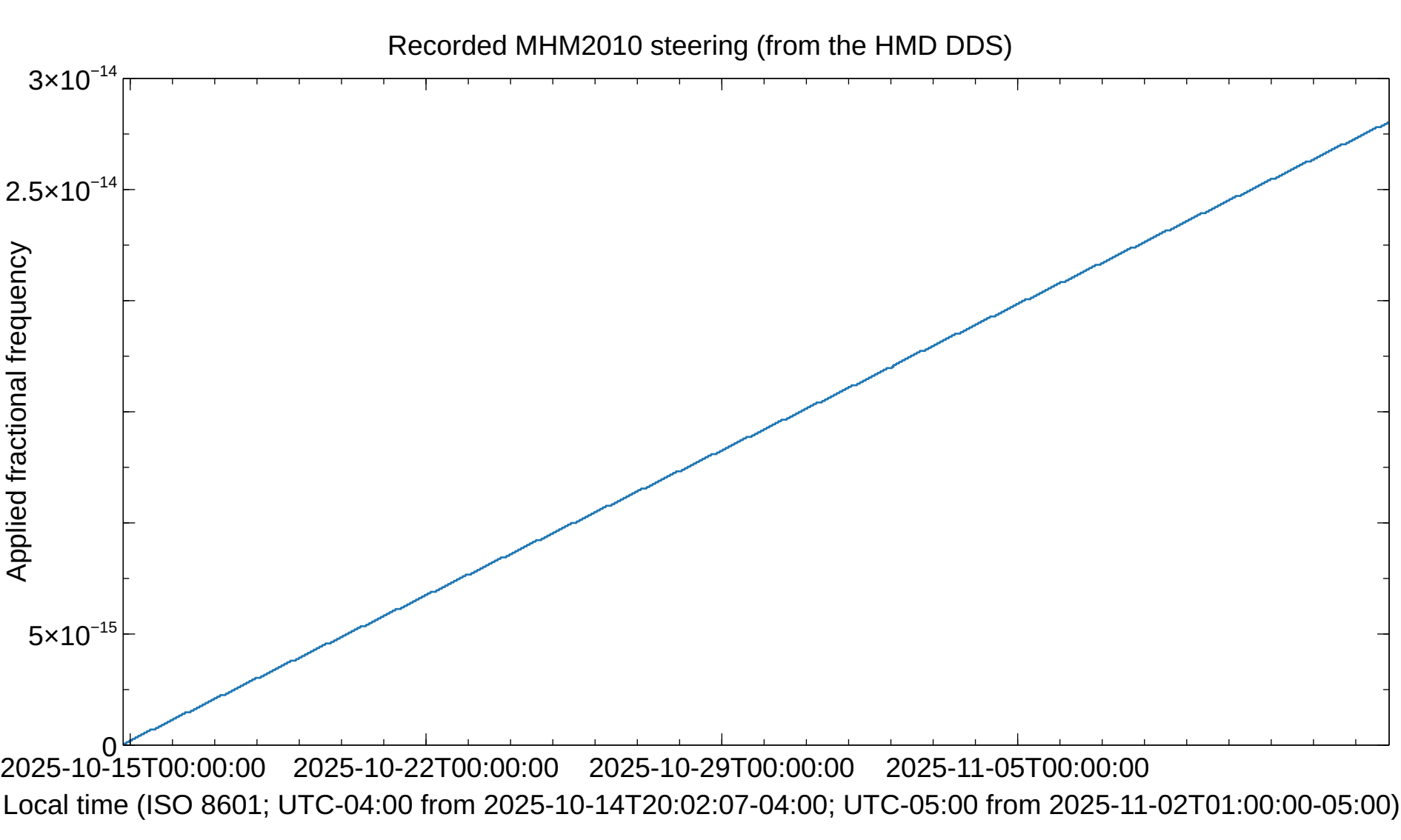

Gap-aware stability, 0.6826894921370859 (one standard deviation) confidence (evaluation; steering removed)
1×10−11
1×10−15
Fractional frequency deviation
100
1000
10000
100000
1000000
Averaging time (s)
OADEV
MDEV
OHDEV

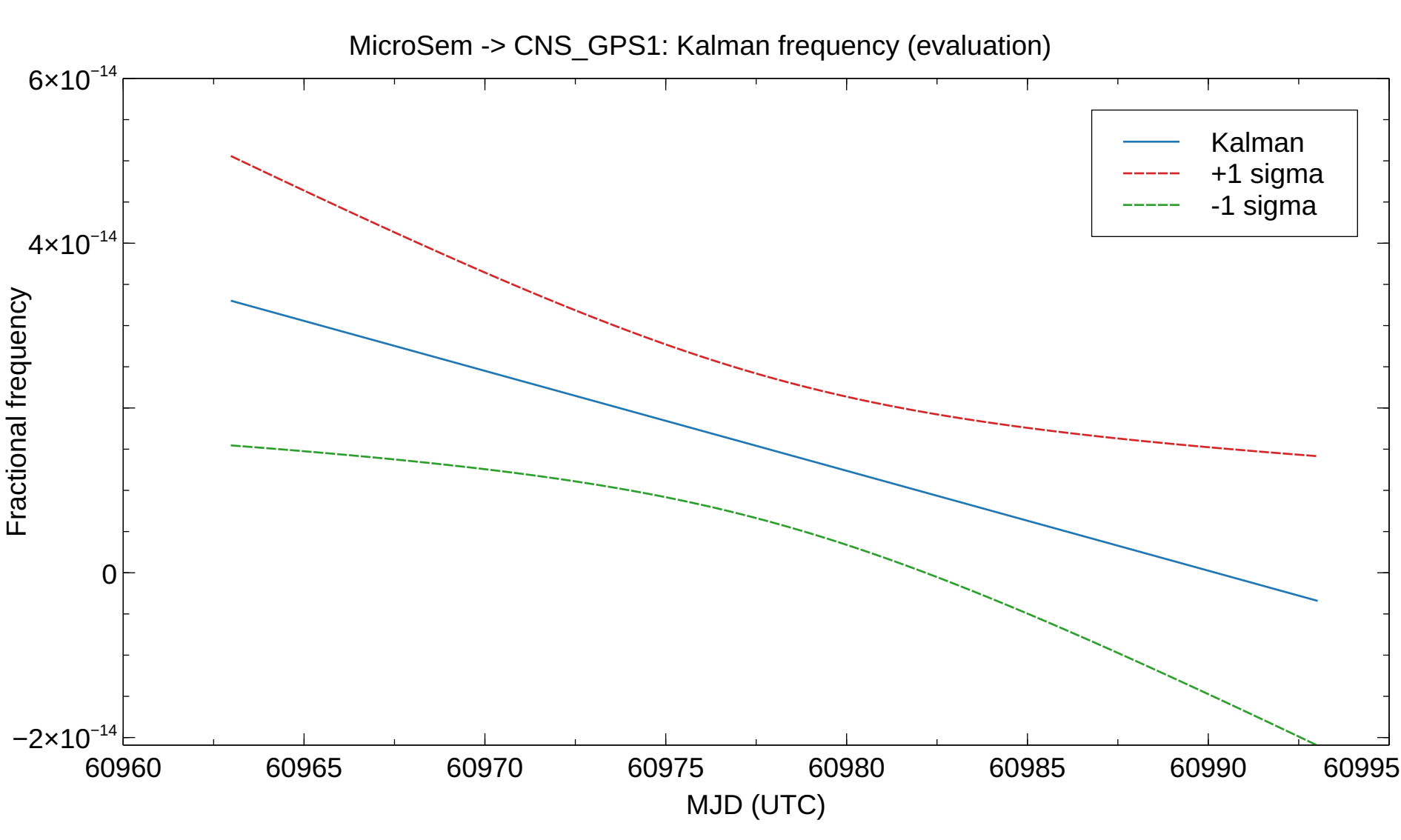

Gap-aware TDEV, 0.6826894921370859 (one standard deviation) confidence (evaluation)
TDEV (s)
2×10−9
1×10−9
5×10−10
100
1000
10000
100000
1000000
Averaging time (s)

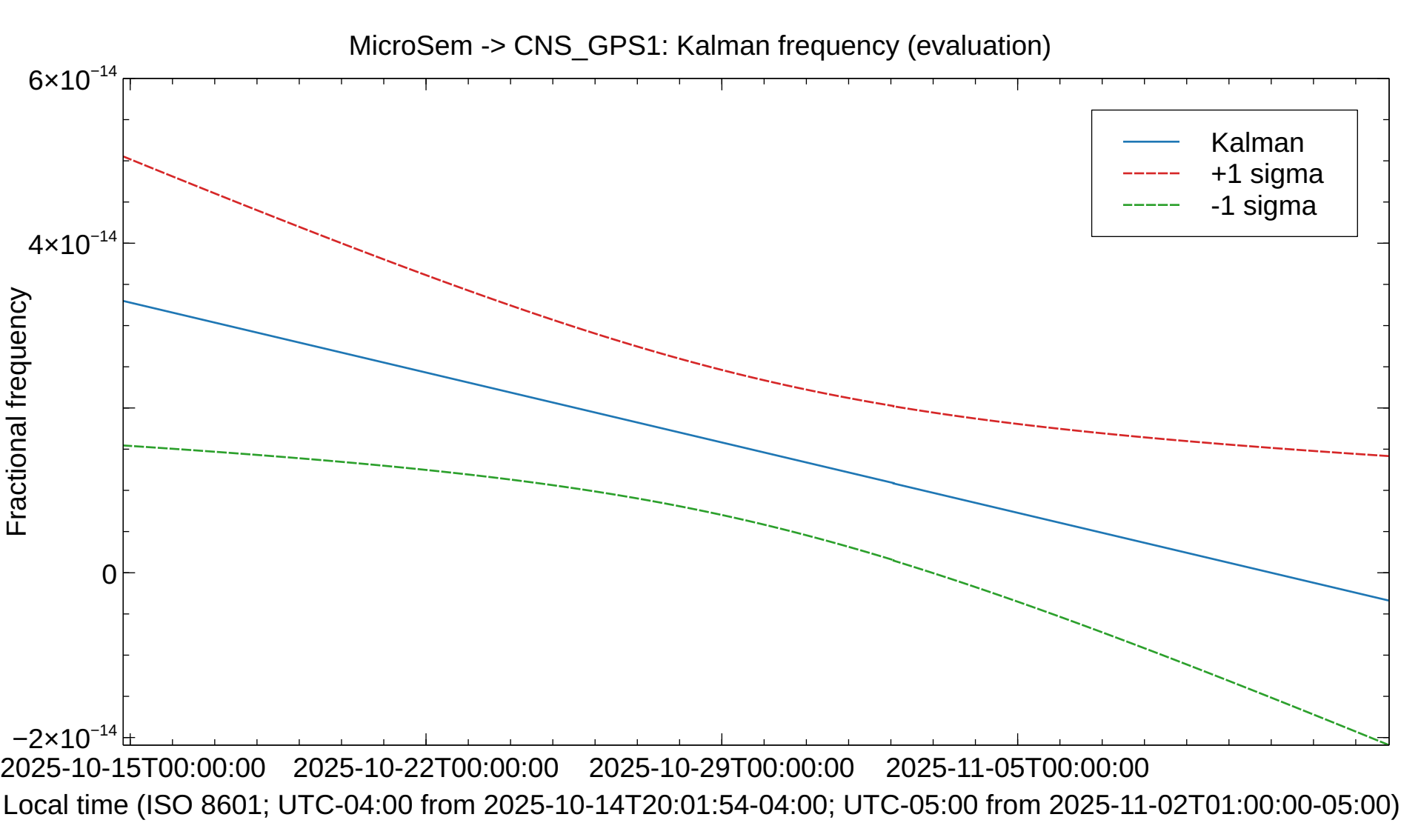

MicroSem -> CNS_GPS1: steering removed (evaluation; mode all)
Measured
Steering removed
Fit of the removed record
Phase (ns)
−2420
−2440
−2460
−2480
−2500
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

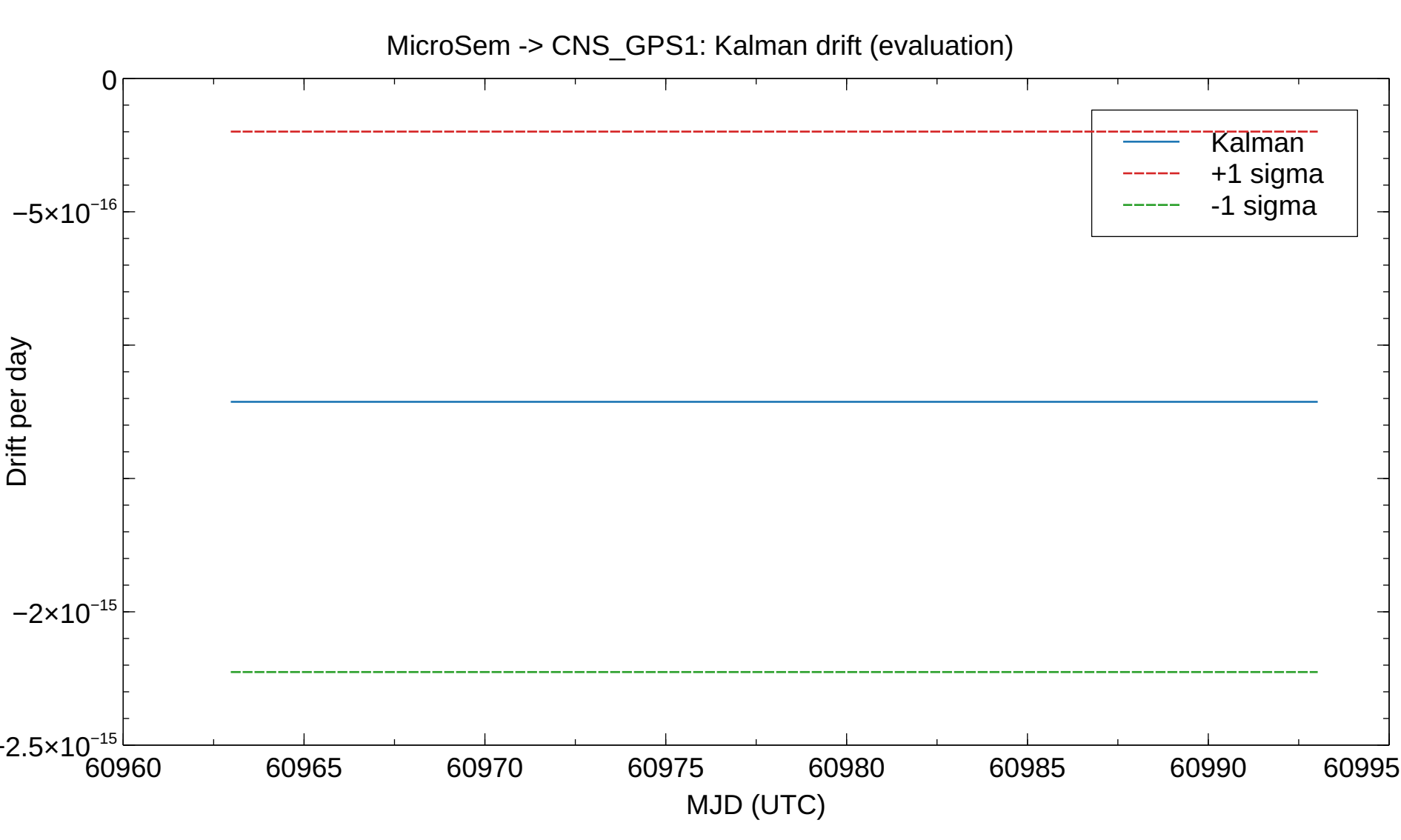

MicroSem -> CNS_GPS1: steering removed (evaluation; mode all)
Measured
Steering removed
Fit of the removed record
Phase (ns)
−2420
−2440
−2460
−2480
−2500
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T17:33:24-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)

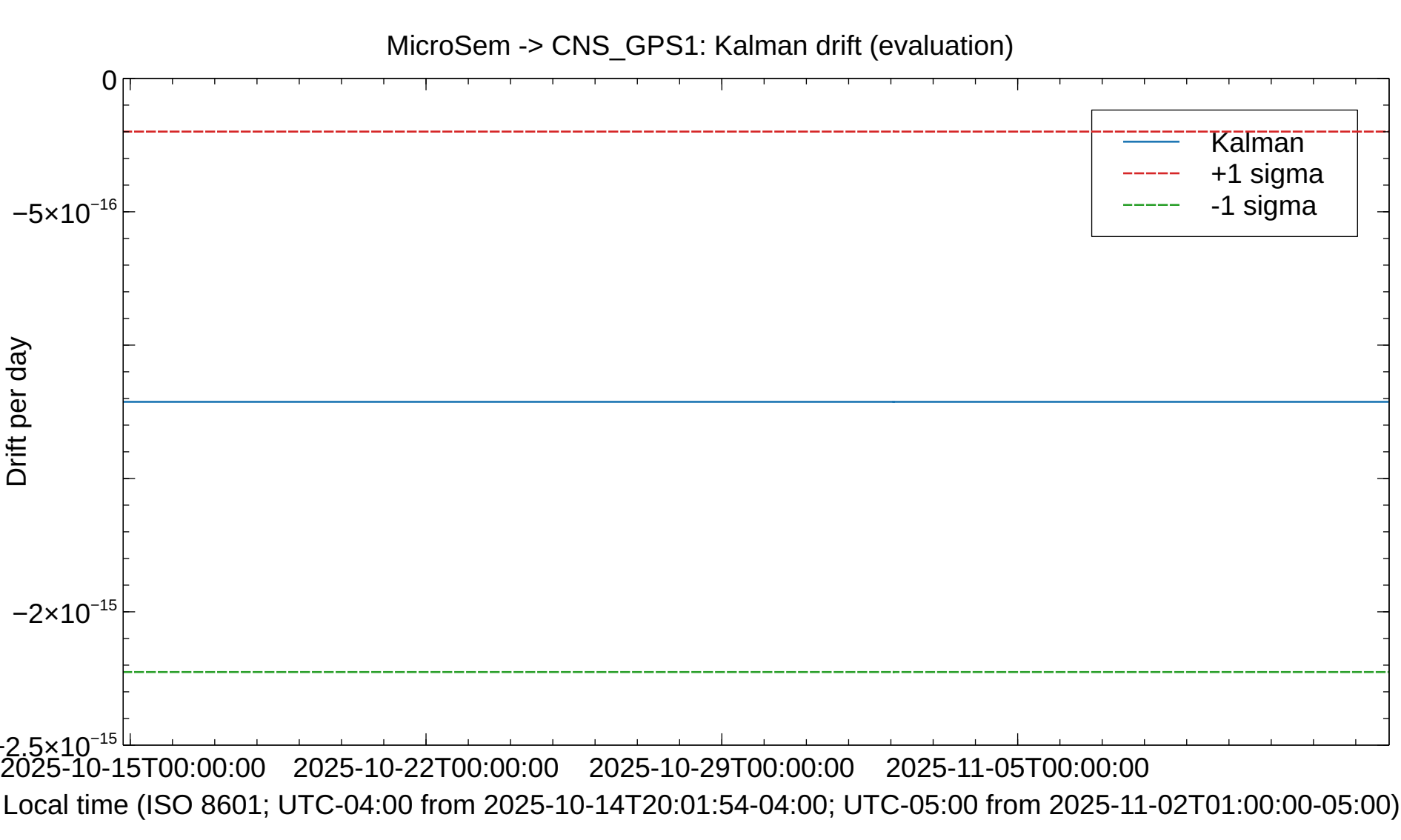

Recorded MHM2010 steering (from the HMD DDS)
3×10−14
2.5×10−14
5×10−15
0
Applied fractional frequency
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

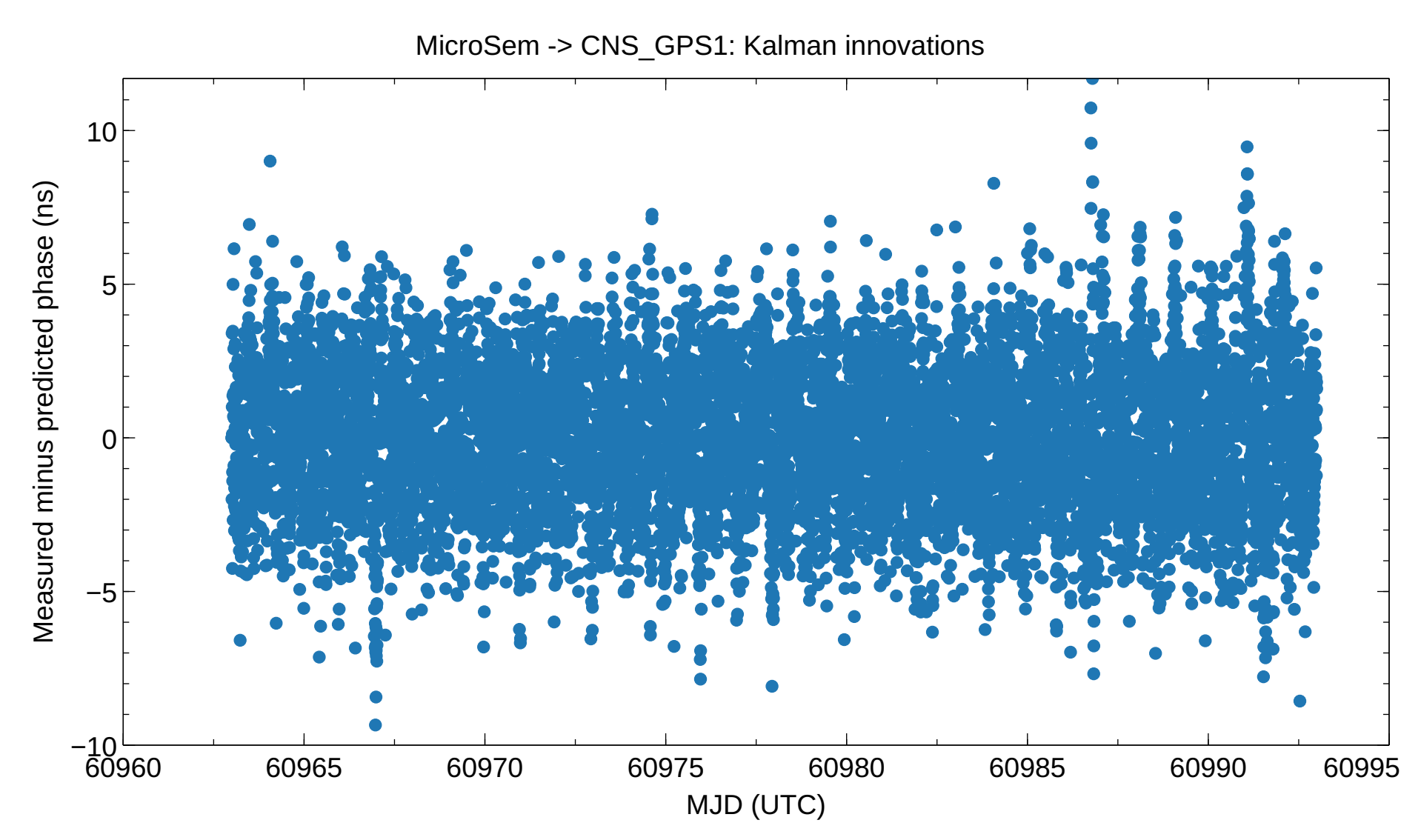

Recorded MHM2010 steering (from the HMD DDS)
3×10−14
2.5×10−14
5×10−15
0
Applied fractional frequency
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:02:07-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)

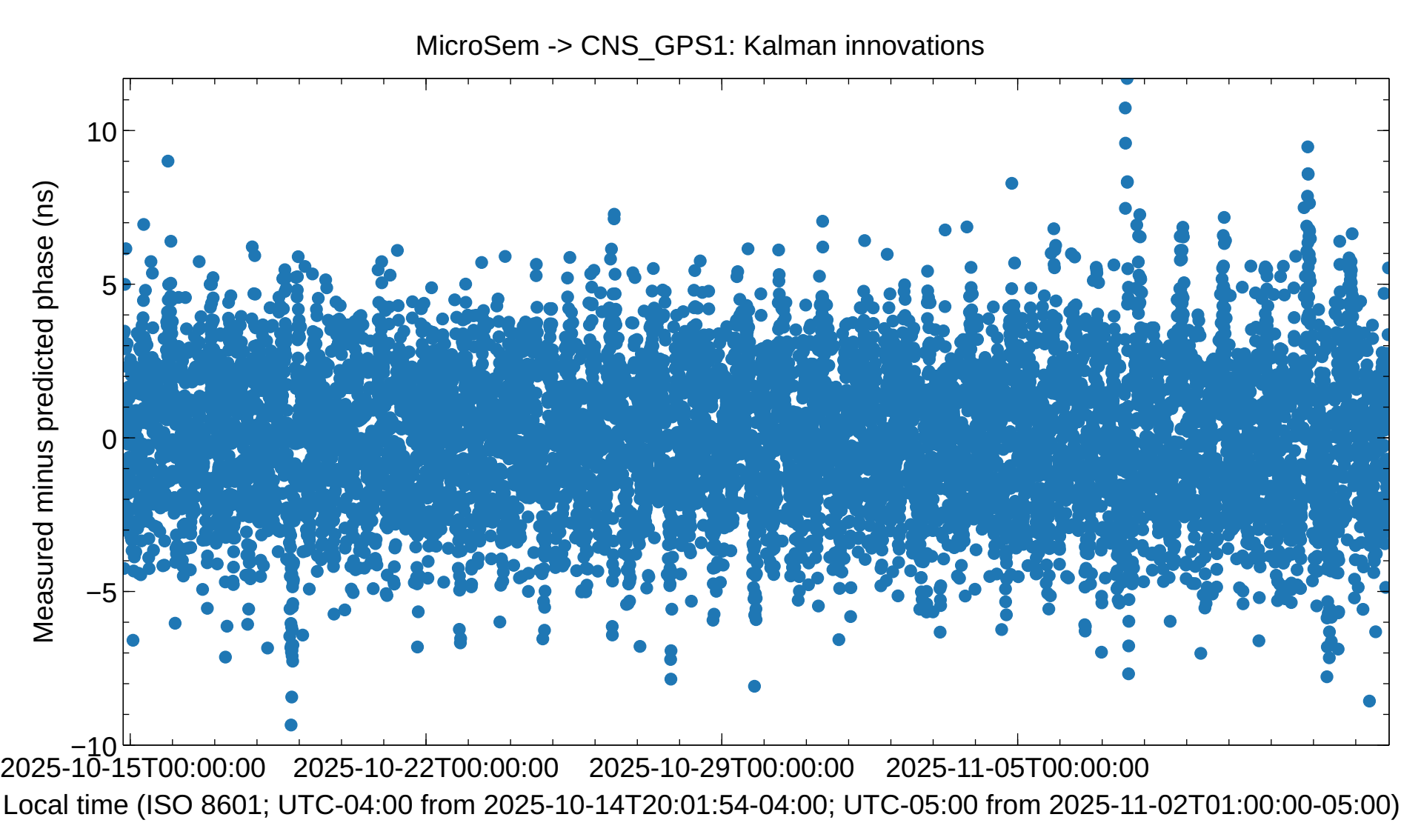

MicroSem -> CNS_GPS1: Kalman frequency (evaluation)
6×10⁻¹⁴
4×10⁻¹⁴
0
−2×10⁻¹⁴
Fractional frequency
Kalman
+1 sigma
-1 sigma
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)

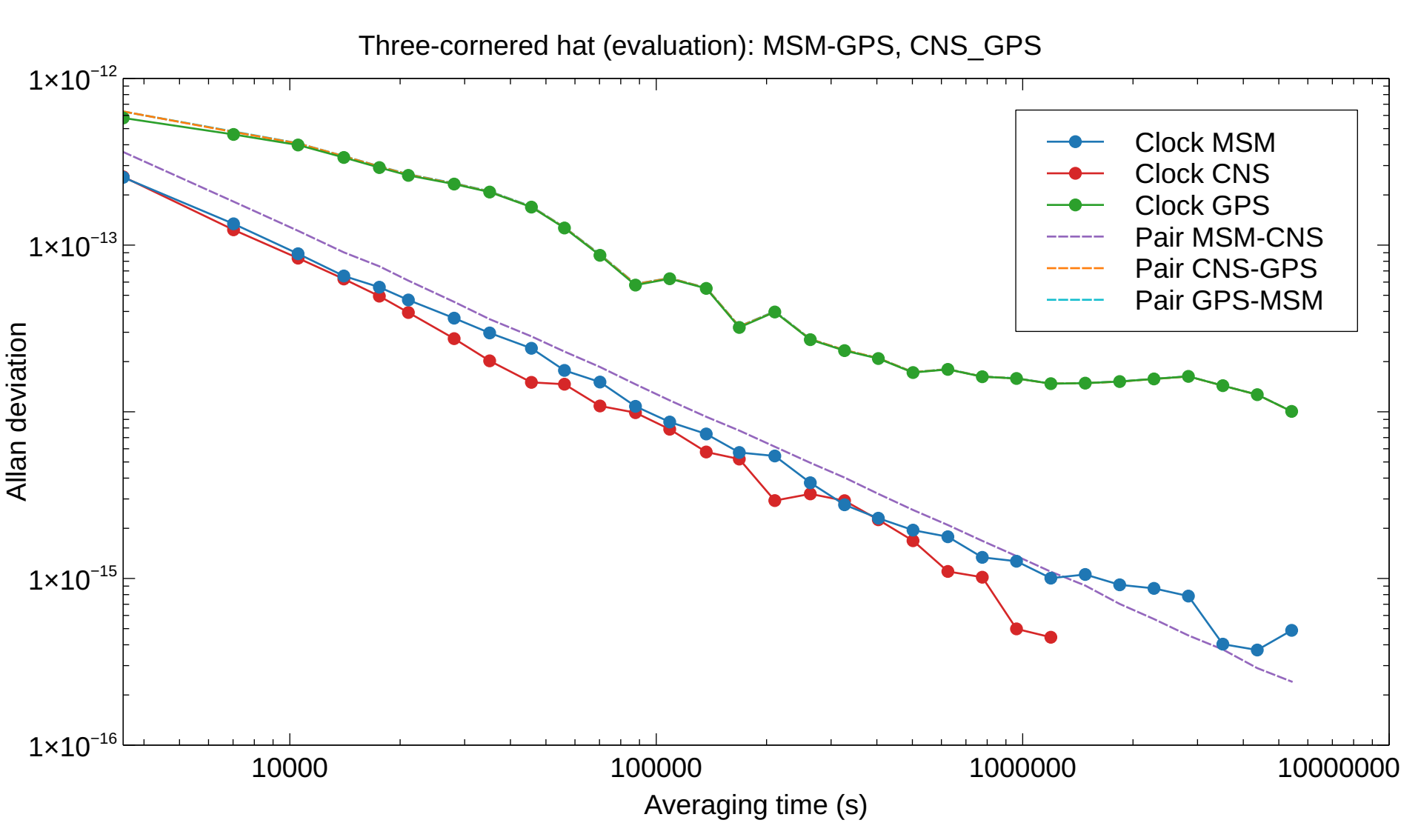

MicroSem -> CNS_GPS1: Kalman frequency (evaluation)
6×10⁻¹⁴
4×10⁻¹⁴
0
−2×10⁻¹⁴
Fractional frequency
Kalman
+1 sigma
-1 sigma
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:01:54-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)


MicroSem -> CNS_GPS1: Kalman drift (evaluation)
0
−5×10−16
−2×10−15
−2.5×10−15
Drift per day
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)
Kalman
+1 sigma
-1 sigma


MicroSem -> CNS_GPS1: Kalman drift (evaluation)
0
−5×10−16
−2×10−15
−2.5×10−15
Drift per day
Kalman
+1 sigma
-1 sigma
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:01:54-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)


MicroSem -> CNS_GPS1: Kalman innovations
Measured minus predicted phase (ns)
10
5
0
−5
−10
60960
60965
60970
60975
60980
60985
60990
60995
MJD (UTC)


MicroSem -> CNS_GPS1: Kalman innovations
Measured minus predicted phase (ns)
10
5
0
−5
−10
2025-10-15T00:00:00
2025-10-22T00:00:00
2025-10-29T00:00:00
2025-11-05T00:00:00
Local time (ISO 8601; UTC-04:00 from 2025-10-14T20:01:54-04:00; UTC-05:00 from 2025-11-02T01:00:00-05:00)


Three-cornered hat (evaluation): MSM-GPS, CNS_GPS
Allan deviation
Averaging time (s)
1×10−12
1×10−13
1×10−15
1×10−16
10000
100000
1000000
10000000
Clock MSM
Clock CNS
Clock GPS
Pair MSM-CNS
Pair CNS-GPS
Pair GPS-MSM